

High-Order Hu–Zhang Mixed Finite Element Methods for Density-Based Topology Optimization

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Abstract. In density-based topology optimization, displacement-based finite elements recover stresses from displacement gradients but generally do not guarantee inter-element traction continuity. Low-order displacement elements are prone to volumetric locking in nearly incompressible regimes. To address these issues, this paper develops a high-order Hu–Zhang mixed finite element framework for density-based topology optimization on simplicial meshes. The framework treats symmetric stress and displacement as independent unknowns. Combining jump stabilization for lower polynomial orders with local corner relaxation, we formulate two optimization models: compliance minimization and volume minimization subject to local stress constraints. We derive consistent adjoint sensitivities that account for the design dependence of the right-hand side introduced by traction lifting. The proposed topology optimization framework is implemented in FEALPy. Two-dimensional numerical examples show that both native higher-order and low-order stabilized formulations exhibit convergence, and the unified optimization workflow is applicable across all investigated discrete polynomial degrees. In standard compliance optimization, the mixed and displacement-based methods yield comparable topologies and compliance values. Mesh convergence results for the fully solid bearing domain show that Hu–Zhang elements exhibit similar compliance convergence trends under compressible and nearly incompressible conditions, without the pronounced loss of accuracy observed for linear displacement elements. In nearly incompressible topology optimization, Hu–Zhang elements effectively mitigate the effects of volumetric locking on the optimization results. The resulting designs are comparable to those obtained with quadratic displacement elements in both topology and compliance evaluated using a common discretization. Local stress constraints are evaluated directly from the primary stress variables without stress recovery, with the discrete stress naturally preserving inter-element normal traction continuity.

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1 Introduction

Topology optimization (TO) provides a systematic numerical design tool for high-performance lightweight structures by optimizing the spatial distribution of material within a prescribed design domain. Since the continuum material distribution approach was introduced by Bendsøe [7], density-based TO, particularly represented by the Solid Isotropic Material with Penalization (SIMP) method, has been widely adopted due to its conceptual clarity, algorithmic maturity, and broad engineering applicability [8]. In classical implementations, the linear elastic state equations are predominantly discretized using standard conforming displacement-based Lagrange finite element methods (LFEM), where stress tensors are subsequently recovered within element interiors via numerical gradients of the discrete displacement fields. While such a displacement-driven paradigm is computationally efficient and sufficient for conventional compliance minimization problems, it encounters severe discretization errors and numerical bottlenecks in design problems governed by accurate local stress evaluation or nearly incompressible material behavior.

In local stress-constrained problems, constraints are imposed element-wise, whose quantity scales with the mesh size, and their values are strictly dictated by the accuracy of the discrete stress field. In standard displacement elements, differentiating displacement shape functions reduces the stress approximation accuracy by one order (e.g., linear elements yield only piecewise constant stresses), accompanied by non-physical discontinuous jumps in both normal and tangential stresses across element interfaces. If these post-processed discrete stresses are directly substituted into local constraints, the mutual coupling among discretization errors, smoothing strategies, and the optimization model may induce local stress oscillations or over-optimized designs [22, 30, 31, 33]. In nearly incompressible material regimes, as Poisson’s ratio approaches the incompressible limit ($\nu_0 \rightarrow 0.5$), the bulk modulus of plane-strain or three-dimensional continua tends to infinity relative to the shear modulus, imposing a stringent divergence-free deformation constraint ($\text{div } \mathbf{u} \approx 0$). Owing to the lack of sufficient deformation degrees of freedom in the discrete displacement space, standard low-order displacement elements suffer from volumetric locking [6], which artificially increases structural stiffness and may distort structural responses and topology evolution [11].

Mixed finite element methods offer an intrinsic variational pathway to overcome the aforementioned mechanical challenges [9]. Based on the Hellinger–Reissner two-field variational principle, the symmetric stress $\boldsymbol{\sigma}$ and displacement $\mathbf{u}$ are treated simultaneously as independent unknowns. The stress trial functions belong directly to the symmetric tensor $H(\text{div})$ space, naturally guaranteeing strict continuity of normal tractions across inter-element boundaries, while the divergence equilibrium equation is satisfied exactly in the weak form. Under parameter-uniform stability conditions, mixed discretizations can avoid volumetric locking in nearly incompressible regimes [9]. The truly-mixed TO studies by Bruggi and Venini [12, 13] and Bruggi [11] demonstrated that directly discretizing stress and displacement provides an effective means for nearly incompressible structural design and stress-constrained optimization. However, most existing mixed finite element TO literature is restricted to lowest-order discretizations, structured quadrilateral meshes, or specific hybridized formulations. For unstructured simplicial meshes (triangles and tetrahedra), constructing strictly symmetric and conforming $H(\text{div})$ spaces has long posed substantial math-

ematical difficulties, often necessitating additional multipliers to relax the symmetry into weakly symmetric formulations [1, 4, 5].

The Hu–Zhang mixed finite element family, introduced by Hu and Zhang [28] and Hu [26], represents a landmark breakthrough in the construction of symmetric stress finite elements. By directly constructing full polynomial spaces $\mathbb{P}_k(\mathcal{S})$ on simplices, this method explicitly decouples normal trace and tangential components via a geometric entity decomposition, ensuring that inter-element normal tractions are uniquely governed by interface degrees of freedom, thereby establishing a family of high-order strongly symmetric conforming spaces for $k \geq d+1$. Building upon this foundation, a suite of essential theories has progressively emerged to facilitate its practical computation: Chen et al. [16] addressed the high polynomial degree requirement of the native formulation ($k \geq d+1$) by devising a symmetric matrix jump stabilization mechanism for lower orders ($k \leq d$); Hu and Ma [27] proposed a local relaxation of stress vertex continuity for incompatible traction data at boundary corners; Chen et al. [17, 18] investigated fast auxiliary space preconditioners and residual-based a posteriori error estimators; and Chen et al. [15] subsequently formulated geometric decompositions and efficient algorithms for high-order simplicial face and edge elements. Incorporating high-order Hu–Zhang mixed finite elements into density-based TO requires a unified variational formulation and computational framework. This framework must account for the design dependence of the discrete right-hand side introduced by non-homogeneous traction lifting and accommodate local relaxation of stress continuity at boundary corners. It must also provide consistent adjoint sensitivities for objectives and constraints expressed in terms of the independent stress variables.

To address the aforementioned theoretical and algorithmic bottlenecks, this paper establishes a high-order Hu–Zhang mixed finite element framework for density-based TO on simplicial meshes, built upon the open-source finite element library FEALPy [37] and the modular TO platform SOPTX [24]. The principal contributions of this work are summarized as follows:

1. **Variational formulation and consistent adjoint analysis for simplicial mixed TO:** Starting from the Hellinger–Reissner variational principle with non-homogeneous Neumann boundary conditions, an extrinsic traction lifting field is employed to handle boundary traction data, systematically revealing the design dependence of the load vector induced by the lifting field. Based on the reconstructed total stress, we derive a consistent adjoint sensitivity analysis framework for both complementary energy objectives and augmented Lagrangian local stress constraints.
2. **Consistent degree-of-freedom management and discrete implementation for optimization iterations:** By casting the high-order symmetric tensor degrees of freedom (DOFs) into a unified management of normal-trace continuity, and combining a common frame convention on shared edges with local relaxation of stress vertex continuity at boundary corners and jump stabilization for lower orders, the stabilized formulations with $k \leq d$ and the native formulations with $k \geq d+1$ run within the same optimization workflow; on a fixed mesh with a fixed stabilization scale, the divergence and stabilization blocks are design-independent, and the material distribution enters the discrete saddle-point system only through the compliance block and the lifted right-hand side.

3. **Cross-order numerical evaluation and mechanical applicability boundaries:** We design a comprehensive benchmark suite spanning low- and high-order manufactured solution convergence rates, cross-order compliance consistency between displacement and mixed formulations, volumetric locking alleviation in nearly incompressible regimes, and denominator-free local stress-constrained optimization. This objectively delineates the practical applicability boundaries of independent symmetric stress approximations relative to standard displacement elements in terms of topological evolution and stress fidelity.

The remainder of this paper is organized as follows. Section 2 introduces the governing equations of linear elasticity, the Hellinger–Reissner mixed variational formulation, and the traction lifting field at the continuum level. Section 3 elaborates on the tensor geometric decomposition on simplices, the construction of stress and displacement finite element spaces, jump stabilization for lower polynomial degrees, local corner stress relaxation, and the assembly of discrete saddle-point systems. Section 4 establishes the two-parameter material interpolation, the complementary energy compliance minimization model, relaxed apparent stress constraints, and consistent adjoint sensitivities based on the total stress. Section 5 presents convergence rate verifications with manufactured solutions alongside three representative benchmark TO problems. Finally, Section 6 summarizes the conclusions and states the scope of the present study.

2 Preliminaries

2.1 Notation and Simplicial Mesh

Let $\Omega \subset \mathbb{R}^d$ ($d = 2, 3$) be a bounded polygonal or polyhedral domain with boundary $\partial\Omega$. The Dirichlet displacement boundary Γ_D and Neumann traction boundary Γ_N are disjoint, relatively open subsets of $\partial\Omega$ such that $\partial\Omega = \overline{\Gamma_D} \cup \overline{\Gamma_N}$. Let $\mathbb{S}$ denote the space of real symmetric second-order tensors of order $d \times d$, and $\mathbf{I}$ be the second-order identity tensor.

For any second-order symmetric tensors $\boldsymbol{\sigma}, \boldsymbol{\tau} \in \mathbb{S}$, the tensor double contraction is defined by $\boldsymbol{\sigma} : \boldsymbol{\tau} = \sum_{i,j=1}^d \sigma_{ij} \tau_{ij} = \text{tr}(\boldsymbol{\sigma} \boldsymbol{\tau})$, and the trace is defined by $\text{tr}(\boldsymbol{\tau}) = \sum_{i=1}^d \tau_{ii}$. The gradient components of a vector field $\mathbf{u}$ are defined by $(\nabla \mathbf{u})_{ij} = \partial u_i / \partial x_j$. For a tensor field $\boldsymbol{\tau}$ and a vector field $\mathbf{u}$, the divergence and symmetric gradient operators are defined by

$$(\text{div } \boldsymbol{\tau})_i = \sum_{j=1}^d \frac{\partial \tau_{ij}}{\partial x_j}, \quad \boldsymbol{\varepsilon}(\mathbf{u}) = \frac{1}{2} \left(\nabla \mathbf{u} + (\nabla \mathbf{u})^\top \right).$$

Let $\mathcal{T}_h$ be a shape-regular conforming simplicial mesh of Ω compatible with the boundary partition, and let $h_T = \text{diam}(T)$, $h = \max_{T \in \mathcal{T}_h} h_T$. Let $\mathcal{V}_h$, $\mathcal{E}_h$, and $\mathcal{F}_h$ denote the sets of vertices, edges, and faces, respectively, where $\mathcal{F}_h = \mathcal{E}_h$ in two dimensions. The face set consists of internal faces $\mathcal{F}_h^i$, Dirichlet boundary faces $\mathcal{F}_h^D$, and Neumann boundary faces $\mathcal{F}_h^N$. For each face $F \in \mathcal{F}_h$, let $h_F = \text{diam}(F)$, and let $|F|$ denote its $(d-1)$ -dimensional measure.

Let $L^2(\Omega; \mathbb{R}^d)$ and $L^2(\Omega; \mathbb{S})$ denote the vector- and symmetric-tensor-valued L^2 spaces, respectively, with their inner products uniformly denoted by $(\cdot, \cdot)_\Omega$. The symmetric tensor $H(\text{div})$

space is defined as

$$H(\operatorname{div}, \Omega; \mathbb{S}) = \left\{ \boldsymbol{\tau} \in L^2(\Omega; \mathbb{S}) : \operatorname{div} \boldsymbol{\tau} \in L^2(\Omega; \mathbb{R}^d) \right\},$$

equipped with the norm $\|\boldsymbol{\tau}\|_{H(\operatorname{div})}^2 = \|\boldsymbol{\tau}\|_{0,\Omega}^2 + \|\operatorname{div} \boldsymbol{\tau}\|_{0,\Omega}^2$.

2.2 Linear Elasticity and Hellinger–Reissner Variational Formulation

Consider the small-strain isotropic linear elasticity boundary value problem:

$$\begin{aligned} -\operatorname{div} \boldsymbol{\sigma} &= \mathbf{b} && \text{in } \Omega, \\ \mathcal{A} \boldsymbol{\sigma} &= \boldsymbol{\varepsilon}(\mathbf{u}) && \text{in } \Omega, \\ \mathbf{u} &= \mathbf{u}_D && \text{on } \Gamma_D, \\ \boldsymbol{\sigma} \mathbf{n} &= \mathbf{g} && \text{on } \Gamma_N, \end{aligned} \tag{2.1}$$

where $\boldsymbol{\sigma} \in \mathbb{S}$ is the symmetric Cauchy stress tensor, $\mathbf{u}$ is the displacement vector, $\mathbf{b}$ is the body force, $\mathbf{g}$ is the prescribed boundary surface traction, and $\boldsymbol{\varepsilon}(\mathbf{u})$ is the small-strain tensor. The fourth-order isotropic linear elastic compliance tensor $\mathcal{A}$ acts on any symmetric second-order tensor $\boldsymbol{\tau}$ as

$$\mathcal{A} \boldsymbol{\tau} = \frac{1}{2\mu} \left(\boldsymbol{\tau} - \frac{\lambda}{2\mu + d\lambda} \operatorname{tr}(\boldsymbol{\tau}) \mathbf{I} \right), \tag{2.2}$$

where λ and μ are the Lamé parameters. Let E and ν denote Young's modulus and Poisson's ratio of the material, respectively. In two dimensions, $\mu = E/[2(1+\nu)]$; $\lambda = E\nu/[(1+\nu)(1-2\nu)]$ for plane strain, and the two-dimensional effective parameter is $\lambda = E\nu/(1-\nu^2)$ for plane stress.

To handle the non-homogeneous Neumann traction boundary condition $\boldsymbol{\sigma} \mathbf{n} = \mathbf{g}$, we introduce the non-homogeneous stress trial set and the homogeneous stress test space:

$$\boldsymbol{\Sigma}_g = \{ \boldsymbol{\tau} \in H(\operatorname{div}, \Omega; \mathbb{S}) : \boldsymbol{\tau} \mathbf{n} = \mathbf{g} \text{ on } \Gamma_N \}, \quad \boldsymbol{\Sigma}_0 = \{ \boldsymbol{\tau} \in H(\operatorname{div}, \Omega; \mathbb{S}) : \boldsymbol{\tau} \mathbf{n} = \mathbf{0} \text{ on } \Gamma_N \},$$

where $\boldsymbol{\Sigma}_0$ is a closed subspace of $H(\operatorname{div}, \Omega; \mathbb{S})$. Assuming $\boldsymbol{\Sigma}_g \neq \emptyset$, it is an affine translation of $\boldsymbol{\Sigma}_0$; when the traction data is non-zero, it does not constitute a linear space. The displacement trial and test spaces are uniformly denoted by $\mathbf{V} = L^2(\Omega; \mathbb{R}^d)$.

The Hellinger–Reissner two-field mixed variational problem for linear elasticity is formulated as: find $(\boldsymbol{\sigma}, \mathbf{u}) \in \boldsymbol{\Sigma}_g \times \mathbf{V}$ such that for all test functions $(\boldsymbol{\tau}, \mathbf{v}) \in \boldsymbol{\Sigma}_0 \times \mathbf{V}$,

$$\begin{aligned} a(\boldsymbol{\sigma}, \boldsymbol{\tau}) + b(\boldsymbol{\tau}, \mathbf{u}) &= \langle \mathbf{u}_D, \boldsymbol{\tau} \mathbf{n} \rangle_{\Gamma_D}, \quad \forall \boldsymbol{\tau} \in \boldsymbol{\Sigma}_0, \\ b(\boldsymbol{\sigma}, \mathbf{v}) &= -(\mathbf{b}, \mathbf{v})_\Omega, \quad \forall \mathbf{v} \in \mathbf{V}, \end{aligned} \tag{2.3}$$

where the bilinear forms are defined by

$$a(\boldsymbol{\sigma}, \boldsymbol{\tau}) = (\mathcal{A} \boldsymbol{\sigma}, \boldsymbol{\tau})_\Omega = \int_\Omega \mathcal{A} \boldsymbol{\sigma} : \boldsymbol{\tau} \, dx, \quad b(\boldsymbol{\tau}, \mathbf{v}) = (\operatorname{div} \boldsymbol{\tau}, \mathbf{v})_\Omega = \int_\Omega (\operatorname{div} \boldsymbol{\tau}) \cdot \mathbf{v} \, dx,$$

and the boundary term $\langle \mathbf{u}_D, \boldsymbol{\tau} \mathbf{n} \rangle_{\Gamma_D}$ denotes the duality pairing between the corresponding trace spaces; when the functions possess sufficient regularity, it can be written as the boundary integral:

$$\langle \mathbf{u}_D, \boldsymbol{\tau} \mathbf{n} \rangle_{\Gamma_D} = \int_{\Gamma_D} \mathbf{u}_D \cdot (\boldsymbol{\tau} \mathbf{n}) \, ds.$$

Since the displacement $\mathbf{u} \in V = L^2(\Omega; \mathbb{R}^d)$, no inter-element continuity is imposed on the displacement field in the variational formulation; the stress belongs to $H(\operatorname{div}, \Omega; \mathbb{S})$ and satisfies inter-element normal traction continuity in the sense of traces. When boundary constraints preclude rigid body motions, the boundary data satisfy compatibility and regularity requirements, and the bilinear forms satisfy continuity, kernel coercivity, and the continuous inf-sup condition, the variational problem (2.3) admits a unique weak solution [9]. For three-dimensional or plane strain problems, in the nearly incompressible limit ($\lambda \rightarrow \infty$ with μ fixed), the compliance bilinear form $a(\cdot, \cdot)$ remains bounded, but along the hydrostatic stress direction, its coercivity constant relative to the L^2 norm tends to zero as $\lambda \rightarrow \infty$. The locking-free property of mixed discretizations must be established under appropriate boundary or mean trace constraints through λ -uniform kernel estimates and discrete inf-sup conditions.

2.3 Non-homogeneous Traction Lifting and Total Stress Formulation

In the Hellinger–Reissner mixed variational formulation, the traction condition $\boldsymbol{\sigma} \mathbf{n} = \mathbf{g}$ is an essential boundary condition constraining the normal trace of stress, whereas the displacement condition is naturally imposed through the boundary functional $\langle \mathbf{u}_D, \boldsymbol{\tau} \mathbf{n} \rangle_{\Gamma_D}$. To handle the non-homogeneous traction condition, we employ the standard lifting argument [21]. Assuming $\boldsymbol{\Sigma}_g \neq \emptyset$, we choose a stress field $\boldsymbol{\sigma}_g \in \boldsymbol{\Sigma}_g$ satisfying the prescribed traction boundary condition as a lifting of the boundary data $\mathbf{g}$. This field does not need to satisfy internal equilibrium equations or constitutive relations. Consequently,

$$\boldsymbol{\Sigma}_g = \boldsymbol{\sigma}_g + \boldsymbol{\Sigma}_0, \quad \boldsymbol{\sigma} = \boldsymbol{\sigma}_0 + \boldsymbol{\sigma}_g, \quad \boldsymbol{\sigma}_0 \in \boldsymbol{\Sigma}_0.$$

Substituting this decomposition into (2.3), we obtain the equivalent mixed variational problem: find $(\boldsymbol{\sigma}_0, \mathbf{u}) \in \boldsymbol{\Sigma}_0 \times V$ such that

$$\begin{aligned} a(\boldsymbol{\sigma}_0, \boldsymbol{\tau}) + b(\boldsymbol{\tau}, \mathbf{u}) &= \langle \mathbf{u}_D, \boldsymbol{\tau} \mathbf{n} \rangle_{\Gamma_D} - a(\boldsymbol{\sigma}_g, \boldsymbol{\tau}), \quad \forall \boldsymbol{\tau} \in \boldsymbol{\Sigma}_0, \\ b(\boldsymbol{\sigma}_0, \mathbf{v}) &= -(\mathbf{b}, \mathbf{v})_\Omega - b(\boldsymbol{\sigma}_g, \mathbf{v}), \quad \forall \mathbf{v} \in V. \end{aligned} \tag{2.4}$$

Once solved, the total stress is reconstructed by $\boldsymbol{\sigma} = \boldsymbol{\sigma}_0 + \boldsymbol{\sigma}_g$. The choice of lifting is not unique: replacing $\boldsymbol{\sigma}_g$ with $\boldsymbol{\sigma}_g + \boldsymbol{\tau}_0$, where $\boldsymbol{\tau}_0 \in \boldsymbol{\Sigma}_0$, alters the homogeneous correction to $\boldsymbol{\sigma}_0 - \boldsymbol{\tau}_0$, while the reconstructed total stress remains invariant.

At the discrete level, we select traction data $\mathbf{g}_h$ belonging to the discrete normal trace space and construct a discrete lifting $\boldsymbol{\sigma}_{g,h}$ in the discrete stress space $\boldsymbol{\Sigma}_{k,h}$ (used in Section 3) satisfying $\boldsymbol{\sigma}_{g,h} \mathbf{n} = \mathbf{g}_h$. The reconstructed discrete total stress exactly satisfies this discrete traction condition, whereas $\mathbf{g} - \mathbf{g}_h$ constitutes the boundary data approximation error. The discrete lifting construction adopted in this work is presented in Section 3.5.

Let ρ denote the material density design variable. In this work, we consider a fixed design domain, a fixed mesh, and design-independent external loads and prescribed displacements, and we fix a choice of lifting σ_g independent of ρ , so that $\partial\sigma_g/\partial\rho=0$. When the compliance operator varies with the design variables, we denote the corresponding bilinear form by $a_\rho(\cdot, \cdot)$. The right-hand side of (2.4) still depends on the design variables through $-a_\rho(\sigma_g, \tau)$, whereas the load functional in the total stress formulation (2.3) is design-independent. Although the two formulations are equivalent, the total stress formulation must retain the affine constraint $\sigma \in \Sigma_g$. In what follows, objective functions and stress constraints are evaluated based on the total stress, and the design dependence of the right-hand side in the lifted formulation is systematically incorporated into the sensitivity derivations.

3 High-Order Hu–Zhang Mixed Finite Elements

3.1 Geometric Tangential–Normal Decomposition on Simplices

Let $T \in \mathcal{T}_h$ be a d -dimensional simplex (a triangle in 2D or a tetrahedron in 3D). Let $\mathbb{P}_k(T; \mathbb{S}) = \mathbb{P}_k(T) \otimes \mathbb{S}$ denote the local symmetric tensor-valued polynomial space of degree at most k ($k \geq 1$) on T , whose dimension is $\dim \mathbb{P}_k(T; \mathbb{S}) = \frac{d(d+1)}{2} \binom{k+d}{d}$.

Hu and Zhang [28] first constructed symmetric $H(\text{div})$ -conforming mixed finite elements on triangular meshes in two dimensions, which was subsequently generalized to arbitrary dimensions and higher-order cases on simplicial meshes by Hu [26]. To systematically organize the stress basis functions and their degrees of freedom (DOFs), we adopt a geometric description that partitions the symmetric tensor components into tangential and normal parts on each sub-simplex. Similar concepts regarding geometric decomposition, local frames, and DOF management can be found in Chen et al. [15] for high-order face and edge elements; below, we extend the geometric decomposition based on tangential–normal frames from vector fields to symmetric tensors.

Let f be an ℓ -dimensional sub-simplex of T ($\ell=0,1,\dots,d$). Let $\{\mathbf{t}_i^f\}_{i=1}^\ell$ be an orthogonal basis for its tangent space, and let $\{\mathbf{n}_j^f\}_{j=1}^{d-\ell}$ be an orthogonal basis for its normal space. Based on this frame, we decompose the symmetric tensor space into a tangential subspace $\mathbb{T}_f(\mathbb{S})$ and a normal subspace $\mathbb{N}_f(\mathbb{S})$:

$$\begin{aligned} \mathbb{T}_f(\mathbb{S}) &= \text{span} \left\{ \text{sym}(\mathbf{t}_i^f \otimes \mathbf{t}_j^f) \right\}_{1 \leq i \leq j \leq \ell}, \\ \mathbb{N}_f(\mathbb{S}) &= \text{span} \left\{ \text{sym}(\mathbf{t}_i^f \otimes \mathbf{n}_j^f) \right\}_{1 \leq i \leq \ell, 1 \leq j \leq d-\ell} \oplus \text{span} \left\{ \text{sym}(\mathbf{n}_i^f \otimes \mathbf{n}_j^f) \right\}_{1 \leq i \leq j \leq d-\ell}, \end{aligned} \quad (3.1)$$

where the symmetric tensor product is defined by $\text{sym}(\mathbf{a} \otimes \mathbf{b}) = \frac{1}{2}(\mathbf{a} \otimes \mathbf{b} + \mathbf{b} \otimes \mathbf{a})$. With respect to the tensor double contraction defined in Section 2.1, the symmetric tensor space $\mathbb{S}$ admits the following orthogonal direct sum decomposition:

$$\mathbb{S} = \mathbb{T}_f(\mathbb{S}) \oplus \mathbb{N}_f(\mathbb{S}).$$

By convention, the span over an empty index set is $\{0\}$. When $\ell = d$, $f = T$, and its normal space is $\{0\}$, which implies $\mathbb{T}_T(\mathbb{S}) = \mathbb{S}$ and $\mathbb{N}_T(\mathbb{S}) = \{0\}$.

$H(\text{div})$ conformity requires that the normal trace of the stress tensor be continuous across the common interface of adjacent elements. In the geometric entity-based construction adopted in this work prior to corner relaxation, the degrees of freedom (DOFs) associated with $\mathbb{N}_f(\mathbb{S})$ are defined under a unified frame and globally shared among elements sharing the sub-simplex f ; this is a constructive approach to achieving normal trace continuity, rather than an obligatory DOF-sharing rule for all conforming stress elements. The local adjustment at corner vertices is detailed in Section 3.4.

For any $\boldsymbol{\tau} \in \mathbb{T}_f(\mathbb{S})$ and any normal vector $\mathbf{n}$ of f , it holds that $\boldsymbol{\tau}\mathbf{n} = \mathbf{0}$. Consequently, although tangential tensor components may be non-zero on the boundary, they do not contribute to tractions along these normal directions. For any boundary face F of element T containing f , its outward unit normal $\mathbf{n}_F$ belongs to the normal space of f , and hence the normal trace of such a component on F vanishes; on boundary faces not containing f , the scalar basis functions associated with the interior nodes of f vanish identically. Therefore, the normal trace of these components vanishes across the entire boundary ∂T , meaning that the corresponding basis functions are $H(\text{div})$ bubble functions on T . Their DOFs are thus not shared among elements sharing f , but are instead numbered locally within each individual element.

Taking a two-dimensional triangular element as an example, the choice of local orthogonal frames and the partition of tensor basis elements on sub-simplices of each dimension are illustrated in Figure 1:

1. When $\ell = 0$ (vertices): $\mathbb{T}_f(\mathbb{S}) = \{0\}$ and $\mathbb{N}_f(\mathbb{S}) = \mathbb{S}$. An arbitrary orthogonal frame can be selected to construct three independent point-value DOFs; for boundary vertices, the frame is aligned with the adjacent boundary edge to facilitate the direct imposition of traction boundary conditions (see Section 3.4 for corners);
2. When $\ell = 1$ (edges): We choose a unit tangent vector $\mathbf{t}_e$ along the edge and a consistently chosen unit normal vector $\mathbf{n}_e$. The normal subspace $\mathbb{N}_e(\mathbb{S})$ consists of $\text{sym}(\mathbf{t}_e \otimes \mathbf{n}_e)$ and $\text{sym}(\mathbf{n}_e \otimes \mathbf{n}_e)$, which determine the normal traction trace components $\boldsymbol{\tau}\mathbf{n}_e$ at interior nodes of the edge and are shared across adjacent elements; the tangential subspace $\mathbb{T}_e(\mathbb{S})$ contains only $\text{sym}(\mathbf{t}_e \otimes \mathbf{t}_e)$ and remains element-private;
3. When $\ell = 2$ (element interior): $\mathbb{T}_T(\mathbb{S}) = \mathbb{S}$ and $\mathbb{N}_T(\mathbb{S}) = \{0\}$, where all interior DOFs are strictly element-private.

A consistent orientation convention is adopted for the tangential and normal vectors of a shared edge between adjacent elements: reversing either $\mathbf{t}_e$ or $\mathbf{n}_e$ flips the sign of the coefficient corresponding to $\text{sym}(\mathbf{t}_e \otimes \mathbf{n}_e)$, whereas the coefficient corresponding to $\text{sym}(\mathbf{n}_e \otimes \mathbf{n}_e)$ remains invariant; therefore, shared DOFs must be defined in a unified frame. When evaluating the element traction associated with its outward normal, an appropriate sign is introduced according to the orientation of the element outward normal relative to the chosen edge normal. In Figure 1, arrows denote the selected local frames (edge normals are not required to point outward from the illustrated element); red markers indicate components shared across adjacent elements, and black markers represent element-private components.

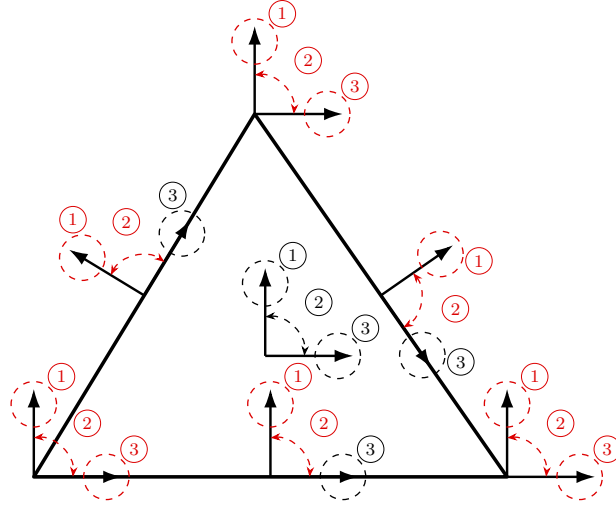


Figure 1: Local orthogonal frames and symmetric tensor components on sub-simplices of a triangular element. Arrows indicate the selected local frame $(\mathbf{a}, \mathbf{b})$; ①, ②, and ③ correspond to $\mathbf{a} \otimes \mathbf{a}$, $\text{sym}(\mathbf{a} \otimes \mathbf{b})$, and $\mathbf{b} \otimes \mathbf{b}$, respectively (on an edge, these correspond to $\mathbf{n}_e \otimes \mathbf{n}_e$, $\text{sym}(\mathbf{t}_e \otimes \mathbf{n}_e)$, and $\mathbf{t}_e \otimes \mathbf{t}_e$); red markers denote components shared across adjacent elements, and black markers represent element-private components.

3.2 Finite Element Spaces and Standard Discrete Variational Formulation

Let the space of continuous piecewise symmetric tensor-valued polynomials and the space of element stress bubble functions be denoted respectively by

$$\tilde{\Sigma}_h^k = \left\{ \boldsymbol{\tau}_h \in H^1(\Omega; \mathbb{S}) : \boldsymbol{\tau}_h|_T \in \mathbb{P}_k(T; \mathbb{S}), \forall T \in \mathcal{T}_h \right\},$$

and

$$\begin{aligned} \mathbf{B}_k(T) &= \{ \boldsymbol{\tau} \in \mathbb{P}_k(T; \mathbb{S}) : \boldsymbol{\tau} \mathbf{n}_T = \mathbf{0} \text{ on } \partial T \}, \\ \mathbf{B}_h^k &= \{ \boldsymbol{\tau}_h \in H(\text{div}, \Omega; \mathbb{S}) : \boldsymbol{\tau}_h|_T \in \mathbf{B}_k(T), \forall T \in \mathcal{T}_h \}, \end{aligned}$$

where $\mathbf{n}_T$ is the unit outward normal of T . The k -th order Hu–Zhang stress finite element space prior to corner relaxation is defined as [16, 26]

$$\Sigma_h^k = \tilde{\Sigma}_h^k + \mathbf{B}_h^k.$$

Since the normal trace of bubble functions vanishes on element boundaries, the space Σ_h^k preserves inter-element traction continuity and retains the entity DOF-sharing rules stipulated in Section 3.1; in particular, all stress components are single-valued at mesh vertices. A local relaxation of this constraint at two-dimensional corner vertices is introduced in Section 3.4.

The corresponding $(k-1)$ -th order piecewise discontinuous displacement finite element space is defined by

$$\mathbf{V}_h^{k-1} = \left\{ \mathbf{v}_h \in L^2(\Omega; \mathbb{R}^d) : \mathbf{v}_h|_T \in \mathbb{P}_{k-1}(T; \mathbb{R}^d), \forall T \in \mathcal{T}_h \right\}.$$

Let $\Sigma_{h,0}^k = \Sigma_h^k \cap \Sigma_0$ denote the homogeneous discrete stress test space. The discrete formulation based on the pair $\Sigma_h^k \times V_h^{k-1}$ is hereafter referred to as the Hu–Zhang mixed finite element method (HZMFEM). Under standard linear elasticity discretization, the discrete mixed variational problem reads: find $(\sigma_{0,h}, \mathbf{u}_h) \in \Sigma_{h,0}^k \times V_h^{k-1}$ such that

$$\begin{aligned} a(\sigma_{0,h}, \boldsymbol{\tau}_h) + b(\boldsymbol{\tau}_h, \mathbf{u}_h) &= \langle \mathbf{u}_D, \boldsymbol{\tau}_h \mathbf{n} \rangle_{\Gamma_D} - a(\sigma_{g,h}, \boldsymbol{\tau}_h), \quad \forall \boldsymbol{\tau}_h \in \Sigma_{h,0}^k, \\ b(\sigma_{0,h}, \mathbf{v}_h) &= -(\mathbf{b}, \mathbf{v}_h)_\Omega - b(\sigma_{g,h}, \mathbf{v}_h), \quad \forall \mathbf{v}_h \in V_h^{k-1}. \end{aligned} \quad (3.2)$$

The discrete total stress is then given by $\sigma_h = \sigma_{0,h} + \sigma_{g,h}$.

According to the theoretical analysis by Hu [26] for the standard problem under pure homogeneous Dirichlet boundary conditions, the Hu–Zhang space pair satisfies the discrete inf-sup condition for high-order schemes with $k \geq d+1$ (i.e., $k \geq 3$ on two-dimensional triangular meshes) under appropriate spatial constructions and theoretical assumptions. Provided that the continuous exact solution possesses sufficient regularity ($\sigma \in H^{k+1}(\Omega; \mathbb{S})$ and $\mathbf{u} \in H^k(\Omega; \mathbb{R}^d)$), the discrete solution to the corresponding standard problem admits the following optimal-order a priori error estimate:

$$\|\sigma - \sigma_h\|_{H(\text{div})} + \|\mathbf{u} - \mathbf{u}_h\|_0 \leq Ch^k (\|\sigma\|_{k+1} + \|\mathbf{u}\|_k). \quad (3.3)$$

In addition, the stress satisfies an optimal-order convergence rate of $\mathcal{O}(h^{k+1})$ in the L^2 norm:

$$\|\sigma - \sigma_h\|_0 \leq Ch^{k+1} \|\sigma\|_{k+1}. \quad (3.4)$$

3.3 Jump Stabilization for Low-Order Elements

For low-order discretizations with $1 \leq k \leq d$ (i.e., $k = 1, 2$ in two dimensions), the discrete inf-sup stability established for standard high-order space pairs is no longer directly applicable, necessitating stabilization of the discrete formulation. To this end, a stabilization mechanism based on symmetric matrix jumps across element faces is introduced on the discontinuous displacement field [16]. For an internal face $F = T^+ \cap T^- \in \mathcal{F}_h^i$ with unit normal vectors $\mathbf{n}^+$ and $\mathbf{n}^-$, respectively, the symmetric matrix jump of a displacement field $\mathbf{v}$ is defined by

$$[[\mathbf{v}]] = \frac{1}{2} (\mathbf{v}^+ \otimes \mathbf{n}^+ + \mathbf{n}^+ \otimes \mathbf{v}^+ + \mathbf{v}^- \otimes \mathbf{n}^- + \mathbf{n}^- \otimes \mathbf{v}^-). \quad (3.5)$$

The matrix-type jump stabilization term employed by Chen et al. [16] takes the form

$$c_{\text{mat},h}(\mathbf{u}_h, \mathbf{v}_h) = \sum_{F \in \mathcal{F}_h} h_F \int_F [[\mathbf{u}_h]] : [[\mathbf{v}_h]] \, ds. \quad (3.6)$$

While the summation in (3.6) is taken over all faces corresponding to the pure displacement boundary conditions in [16], for the mixed boundary conditions considered in this work, the faces involved in the penalty are taken as $\mathcal{F}_h^i \cup \mathcal{F}_h^D$. On the Dirichlet boundary Γ_D , the one-sided jump is defined as $[[\mathbf{v}]] = \text{sym}(\mathbf{v} \otimes \mathbf{n})$, and the penalty is applied to the displacement deviation $\mathbf{u}_h - \mathbf{u}_D$.

When dimensional variables are involved, we further perform dimensional matching for the matrix-type jump term. Compared with the bilinear form $b(\cdot, \cdot)$ in the mixed variational formulation, the unscaled jump term carries an extra dimension of length squared while lacking the dimension of stress. Therefore, we utilize the shear modulus μ and a fixed macroscopic characteristic length L_0 to construct the scaling parameter $\alpha = \mu / L_0^2$, yielding

$$\alpha = \frac{\mu}{L_0^2}, \quad c_h(\mathbf{u}_h, \mathbf{v}_h) = \sum_{F \in \mathcal{F}_h^i \cup \mathcal{F}_h^D} \alpha h_F \int_F [\![\mathbf{u}_h]\!] : [\![\mathbf{v}_h]\!] \, ds, \quad (3.7)$$

where μ is taken as the shear modulus of the solid material, fixed at $E_0 / [2(1 + \nu_0)]$ throughout the TO process without being updated by material density interpolation; L_0 is chosen as the maximum edge length of the bounding box of the fixed domain mesh, namely $L_0 = \max_{1 \leq i \leq d} (x_i^{\max} - x_i^{\min})$. Consequently, α is strictly design-independent, and the stabilization matrix generates no additional sensitivity terms under a fixed mesh and invariant boundary partition.

The known boundary functional corresponding to non-homogeneous Dirichlet boundary conditions adopts the identical scaling:

$$\ell_h^D(\mathbf{v}_h) = \sum_{F \in \mathcal{F}_h^D} \alpha h_F \int_F [\![\mathbf{u}_D]\!] : [\![\mathbf{v}_h]\!] \, ds. \quad (3.8)$$

Incorporating the stabilization terms, the unified discrete mixed variational problem is formulated as follows: find $(\sigma_{0,h}, \mathbf{u}_h) \in \Sigma_{h,0}^k \times \mathbf{V}_h^{k-1}$ such that

$$\begin{aligned} a(\sigma_{0,h}, \boldsymbol{\tau}_h) + b(\boldsymbol{\tau}_h, \mathbf{u}_h) &= \langle \mathbf{u}_D, \boldsymbol{\tau}_h \mathbf{n} \rangle_{\Gamma_D} - a(\sigma_{g,h}, \boldsymbol{\tau}_h), & \forall \boldsymbol{\tau}_h \in \Sigma_{h,0}^k, \\ b(\sigma_{0,h}, \mathbf{v}_h) - c_h(\mathbf{u}_h, \mathbf{v}_h) &= -(\mathbf{b}, \mathbf{v}_h)_\Omega - b(\sigma_{g,h}, \mathbf{v}_h) - \ell_h^D(\mathbf{v}_h), & \forall \mathbf{v}_h \in \mathbf{V}_h^{k-1}, \end{aligned} \quad (3.9)$$

where the stabilization terms are activated for low-order discretizations with $1 \leq k \leq d$, while for high-order discretizations with $k \geq d+1$, one sets $c_h = 0$ and $\ell_h^D = 0$ so that (3.9) naturally reduces to the standard mixed formulation (3.2). In two dimensions, these two regimes correspond to $k=1, 2$ and $k \geq 3$, respectively. In addition, the stabilization terms are imposed exclusively on interior faces and the Dirichlet boundary; the traction boundary condition on Γ_N is enforced through the stress normal trace in the variational formulation, and thus no displacement jump penalty is introduced on Γ_N .

Regarding theoretical error estimates for the low-order stabilized formulation, Chen et al. [16] established the discrete inf-sup condition for the original stabilized scheme ($1 \leq k \leq d$) under pure Dirichlet boundary conditions. Provided that the exact solution possesses sufficient regularity ($\sigma \in H^{k+1}(\Omega; \mathbb{S})$ and $\mathbf{u} \in H^k(\Omega; \mathbb{R}^d)$), the discrete solution satisfies the a priori error estimate:

$$\|\sigma - \sigma_h\|_{H(\text{div}, A)} + \|\mathbf{u} - \mathbf{u}_h\|_{0,c} \leq Ch^k (\|\sigma\|_{k+1} + \|\mathbf{u}\|_k), \quad (3.10)$$

where the compliance-weighted stress norm is defined by $\|\boldsymbol{\tau}\|_{H(\text{div}, A)}^2 = a(\boldsymbol{\tau}, \boldsymbol{\tau}) + \|\text{div } \boldsymbol{\tau}\|_0^2$, and the mesh-dependent displacement norm is defined by $\|\mathbf{v}\|_{0,c}^2 = \|\mathbf{v}\|_0^2 + c_{\text{mat},h}(\mathbf{v}, \mathbf{v})$. Under the mean trace elimination condition $\int_\Omega \text{tr } \boldsymbol{\tau} \, dx = 0$, this weighted norm is uniformly equivalent to the standard $H(\text{div})$ norm with respect to the Lamé parameter λ . The theoretical stress error estimate in the L^2 norm for this low-order scheme is of order $\mathcal{O}(h^k)$, which is one order lower than the optimal estimate for the standard high-order formulation.

3.4 Partial Relaxation of Vertex Continuity at Corners

At a corner vertex x_c of a two-dimensional polygonal domain, let e_+ and e_- denote two adjacent traction boundary edges meeting at x_c , with unit outward normal vectors $\mathbf{n}_+$ and $\mathbf{n}_-$, and prescribed tractions $\mathbf{g}_+$ and $\mathbf{g}_-$, respectively. If the stress at the corner is represented by a single symmetric tensor, it must satisfy the compatibility condition $\mathbf{n}_- \cdot \mathbf{g}_+(x_c) = \mathbf{n}_+ \cdot \mathbf{g}_-(x_c)$. When the prescribed boundary data do not satisfy this compatibility condition, the constraint of a single-valued stress tensor at the vertex impedes the accurate enforcement of traction boundary conditions on both sides [27].

To resolve this discrete constraint conflict, we adopt the partial relaxation of vertex continuity technique [27]. In the following, we restrict the discussion to the two-element corner configuration in two dimensions: the neighborhood of the corner consists of two triangular elements T^+ and T^- , whose common edge e connects the corner vertex to the interior of the domain; if the corner initially belongs to only a single triangle, that triangle can be subdivided into two triangles beforehand. While preserving the continuity of the normal trace across the common edge e , the pure tangential stress components of the adjacent elements relative to e are decoupled. Letting $\mathcal{V}_c$ denote the set of corner vertices subjected to this relaxation, the corresponding relaxed discrete stress space is defined as

$$\begin{aligned} \Sigma_{h,\text{rel}}^k = \{ & \boldsymbol{\tau}_h \in H(\text{div}, \Omega; \mathbb{S}) : \boldsymbol{\tau}_h|_T \in \mathbb{P}_k(T; \mathbb{S}), \quad \forall T \in \mathcal{T}_h, \\ & \boldsymbol{\tau}_h \text{ is single-valued at each vertex in } \mathcal{V}_h \setminus \mathcal{V}_c \}. \end{aligned} \quad (3.11)$$

For the two-element configuration described above, the two traction components in the frame of the common edge remain shared, while the pure tangential stress components are decoupled across the two sides, thereby increasing the number of global independent degrees of freedom at the corner from 3 to 4. The detailed implementation procedure is illustrated in Figure 2:

1. *Common edge frame*: Choose a unified orthonormal frame comprising the unit tangential vector $\mathbf{t}_e$ and normal vector $\mathbf{n}_e$ for the common edge e ;
2. *Tangential component splitting*: Split the vertex degree of freedom corresponding to $\mathbf{t}_e \otimes \mathbf{t}_e$ into two independent degrees of freedom belonging to T^+ and T^- , respectively;
3. *Normal trace preservation*: Keep the components corresponding to $\mathbf{n}_e \otimes \mathbf{n}_e$ and $\text{sym}(\mathbf{t}_e \otimes \mathbf{n}_e)$ shared between T^+ and T^- , while the remaining normal trace degrees of freedom on the common edge are assembled according to standard rules;
4. *Boundary condition enforcement*: Through a local 4×4 basis transformation at the corner, transform the four degrees of freedom into two traction components on each of the two boundary edges e_+ and e_- (corresponding to the corner basis functions of Hu and Ma [27]), thereby allowing traction conditions to be applied directly edge by edge; this transformation is invertible whenever e_+ and e_- are not collinear. We emphasize that this procedure only increases the number of global independent degrees of freedom in the corner neighborhood, without altering the local polynomial space on each triangle.

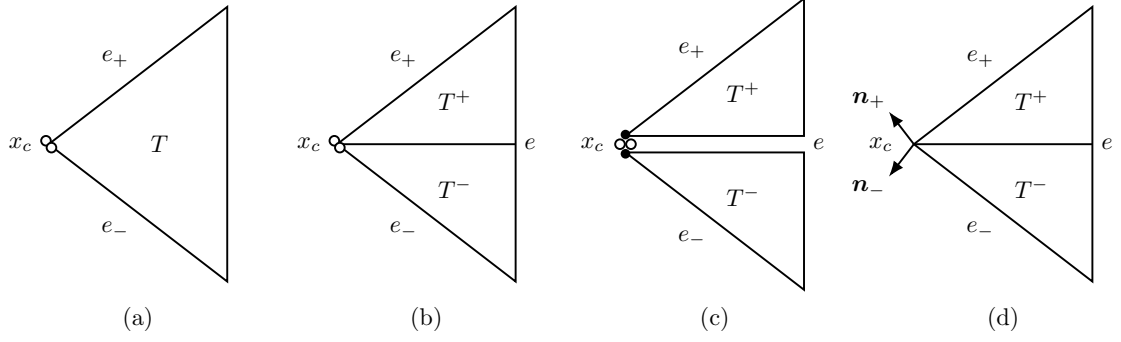


Figure 2: Partial relaxation of vertex stress continuity at a two-dimensional corner: (a) single-element configuration; (b) subdivision into two triangles sharing edge e ; (c) independent tangential degrees of freedom (solid dots) and shared normal-trace degrees of freedom (open circles); (d) boundary traction representation after a local basis transformation. Arrows indicate the outward unit normals $n_{\pm}$. The gap in (c) is schematic only.

For any polynomial degree $k \geq 1$, the local symmetric stress space on a two-dimensional triangle has dimension

$$\dim \mathbb{P}_k(T; \mathbb{S}) = \frac{3(k+1)(k+2)}{2}.$$

The degrees of freedom are grouped by geometric entity as follows:

- *Vertex DOFs*: each element is assigned 3 local components at each vertex; components at unrelaxed vertices are shared globally, whereas the aforementioned two-element relaxed corner contributes a total of 4 global independent degrees of freedom;
- *Edge interior DOFs*: each edge is assigned $2(k-1)$ normal traction degrees of freedom (shared across adjacent elements) and $(k-1)$ tangential normal stress degrees of freedom (strictly element-private);
- *Element interior DOFs*: each element contains $\frac{3(k-1)(k-2)}{2}$ internal stress degrees of freedom, which are strictly element-private.

In two dimensions, all subsequent discrete variational problems and algebraic matrix assemblies adopt the relaxed stress space constructed above. For notational simplicity, we continue to denote this relaxed space by Σ_h^k , and the corresponding discrete stress space satisfying homogeneous traction boundary conditions by $\Sigma_{h,0}^k = \Sigma_h^k \cap \Sigma_0$.

3.5 Discrete Saddle-Point System

Let $\{\Phi_i\}_{i=1}^{N_\sigma}$ denote the global basis functions of the homogeneous discrete stress space $\Sigma_{h,0}^k$, and let $\{\psi_p\}_{p=1}^{N_u}$ denote the global basis functions of the piecewise discontinuous displacement space

$\mathbf{V}_h^{k-1}$. The discrete solutions are expanded as

$$\boldsymbol{\sigma}_{0,h} = \sum_{j=1}^{N_\sigma} s_{0,j} \boldsymbol{\Phi}_j, \quad \mathbf{u}_h = \sum_{q=1}^{N_u} u_q \boldsymbol{\psi}_q.$$

Substituting these basis expansions into the unified discrete mixed formulation (3.9) leads to the following symmetric indefinite discrete saddle-point algebraic system:

$$\begin{bmatrix} \mathbf{A} & \mathbf{B} \\ \mathbf{B}^\top & -\mathbf{C} \end{bmatrix} \begin{bmatrix} \mathbf{s}_0 \\ \mathbf{u} \end{bmatrix} = \begin{bmatrix} \mathbf{f}_\sigma \\ \mathbf{f}_u \end{bmatrix}, \quad (3.12)$$

where the entries of each block matrix are explicitly given by:

$$\begin{aligned} A_{ij} &= a(\boldsymbol{\Phi}_j, \boldsymbol{\Phi}_i) = \int_{\Omega} \mathcal{A} \boldsymbol{\Phi}_j : \boldsymbol{\Phi}_i \, dx, & i, j &= 1, \dots, N_\sigma, \\ B_{ip} &= b(\boldsymbol{\Phi}_i, \boldsymbol{\psi}_p) = \int_{\Omega} (\operatorname{div} \boldsymbol{\Phi}_i) \cdot \boldsymbol{\psi}_p \, dx, & i &= 1, \dots, N_\sigma; \, p = 1, \dots, N_u, \\ C_{pq} &= c_h(\boldsymbol{\psi}_q, \boldsymbol{\psi}_p) = \sum_{F \in \mathcal{F}_h^i \cup \mathcal{F}_h^D} \alpha h_F \int_F \llbracket \boldsymbol{\psi}_q \rrbracket : \llbracket \boldsymbol{\psi}_p \rrbracket \, ds, & p, q &= 1, \dots, N_u. \end{aligned} \quad (3.13)$$

The components of the right-hand side load vectors are defined by:

$$\begin{aligned} (f_\sigma)_i &= \int_{\Gamma_D} \mathbf{u}_D \cdot (\boldsymbol{\Phi}_i \mathbf{n}) \, ds - \int_{\Omega} \mathcal{A} \boldsymbol{\sigma}_{g,h} : \boldsymbol{\Phi}_i \, dx, & i &= 1, \dots, N_\sigma, \\ (f_u)_p &= - \int_{\Omega} \mathbf{b} \cdot \boldsymbol{\psi}_p \, dx - \int_{\Omega} (\operatorname{div} \boldsymbol{\sigma}_{g,h}) \cdot \boldsymbol{\psi}_p \, dx - \ell_h^D(\boldsymbol{\psi}_p), & p &= 1, \dots, N_u, \end{aligned} \quad (3.14)$$

where $\mathbf{s}_0 = (s_{0,1}, \dots, s_{0,N_\sigma})^\top$ is the vector of unknown degrees of freedom for the homogeneous stress, and $\mathbf{u} = (u_1, \dots, u_{N_u})^\top$ is the displacement degree-of-freedom vector. Degrees of freedom at boundary vertices and boundary edges are defined with respect to local frames aligned with the boundary (with corner vertices treated as described in Section 3.4), so that the normal stress trace on Γ_N is directly represented by an identified subset of degrees of freedom. By interpolating the traction data $\mathbf{g}$ at $k+1$ equidistant Lagrange nodes on each traction boundary edge to obtain $\mathbf{g}_h$, assigning these values to the corresponding boundary degrees of freedom, and setting all remaining degrees of freedom to zero, one obtains the discrete lifting $\boldsymbol{\sigma}_{g,h} \in \boldsymbol{\Sigma}_h^k$ satisfying $\boldsymbol{\sigma}_{g,h} \mathbf{n} = \mathbf{g}_h$. Accordingly, among the global basis functions of the full space $\boldsymbol{\Sigma}_h^k$ determined by the degrees of freedom in Sections 3.1 and 3.4, eliminating those constrained degrees of freedom on Γ_N directly yields the basis functions $\{\boldsymbol{\Phi}_i\}_{i=1}^{N_\sigma}$ for the homogeneous subspace $\boldsymbol{\Sigma}_{h,0}^k$.

4 Density-Based Topology Optimization with Independent Stress

4.1 Design Variables, Density Filter, Projection, and Dual-Parameter Material Interpolation

The design domain $\Omega \subset \mathbb{R}^d$ is discretized into a finite element mesh $\mathcal{T}_h = \{T_e\}_{e=1}^{N_e}$ consisting of N_e elements. In the density-based framework for continuum TO, each element T_e carries a

mathematical design variable $\rho_e \in [0,1]$, and these variables form the global design vector $\boldsymbol{\rho} = (\rho_1, \dots, \rho_{N_e})^\top$.

To alleviate mesh dependency and checkerboard patterns, and to introduce a spatial smoothing length scale, a linear density filter [10, 14] maps the design variables ρ_e to the filtered element densities $\tilde{\rho}_e$:

$$\tilde{\rho}_e = \frac{\sum_{j \in \mathcal{N}_e} w_{ej} v_j \rho_j}{\sum_{j \in \mathcal{N}_e} w_{ej} v_j}, \quad (4.1)$$

where $v_j = |T_j|$ is the measure of element T_j (area in two dimensions, volume in three dimensions), and $\mathcal{N}_e = \{j \in \{1, \dots, N_e\} : \|\mathbf{x}_e - \mathbf{x}_j\|_2 \leq r_{\min}\}$ is the index set of the elements whose centroids lie within the filter radius $r_{\min}$ of the centroid $\mathbf{x}_e = \frac{1}{|T_e|} \int_{T_e} \mathbf{x} d\mathbf{x}$ of T_e . The linearly decaying convolution weights w_{ej} are defined by

$$w_{ej} = \max(0, r_{\min} - \|\mathbf{x}_e - \mathbf{x}_j\|_2).$$

To reduce the intermediate densities produced by filtering and to sharpen the material boundaries, we adopt the smoothed Heaviside projection of [36], which leads to a three-field representation in terms of the design, filtered, and physical densities:

$$\boldsymbol{\rho} \xrightarrow{\text{density filter}} \tilde{\boldsymbol{\rho}} \xrightarrow{\text{smoothed projection}} \bar{\boldsymbol{\rho}}, \quad \bar{\rho}_e = P_{\beta, \eta_p}(\tilde{\rho}_e) = \frac{\tanh(\beta \eta_p) + \tanh[\beta(\tilde{\rho}_e - \eta_p)]}{\tanh(\beta \eta_p) + \tanh[\beta(1 - \eta_p)]}, \quad (4.2)$$

where $\eta_p \in (0,1)$ is the projection threshold and $\beta > 0$ controls the sharpness of the projection; η_p is distinct from the stress threshold function $\eta(\cdot)$ introduced later. When the projection is enabled, β is increased in stages to sharpen the boundaries; when it is disabled, the identity map $\bar{\rho}_e = \tilde{\rho}_e$ is used. The subsequent material interpolation, state equations, volume evaluation, and local stress constraints are all formulated in terms of the physical density $\bar{\rho}_e$.

If Poisson's ratio is held fixed and only Young's modulus is interpolated, elements of low and intermediate density remain nearly incompressible as $\nu_0 \rightarrow 0.5$. They then behave as a fluid-like medium with very low shear resistance but very high resistance to volumetric deformation, which can transmit hydrostatic pressure and be exploited by the optimizer. To make the low- and intermediate-density phase behave as a weak compressible material, we interpolate Young's modulus and Poisson's ratio separately as functions of density (see [12]):

1. Young's modulus interpolation (SIMP):

$$E(\bar{\rho}_e) = E_{\min} + (E_0 - E_{\min}) \bar{\rho}_e^{p_E}, \quad (4.3)$$

where E_0 is Young's modulus of the isotropic solid material and $E_{\min} > 0$ is a lower bound on the modulus that prevents the compliance operator from becoming undefined at zero modulus; its value is specified for each example. The exponent p_E penalizes intermediate stiffness; we take $p_E = 3$ for the compliance examples and $p_E = 3.5$ for the stress-constrained examples.

2. Poisson's ratio relaxation:

$$\nu(\bar{\rho}_e) = \nu_{\text{void}} + (\nu_0 - \nu_{\text{void}}) \bar{\rho}_e^{p_\nu}, \quad (4.4)$$

where ν_0 is the target Poisson's ratio of the solid material (e.g., $\nu_0 = 0.4999$) and ν_{void} is the reference Poisson's ratio of the void region. We take $\nu_{\text{void}} = 0.3$ and the relaxation exponent $p_\nu = 1$, so that Poisson's ratio varies linearly with the physical density from the void reference value to the base-material value. This keeps the low- and intermediate-density phase away from the incompressible limit and removes the fluid-like load-carrying mechanism described above. Equation (4.4) is activated for the nearly incompressible optimization group with $\nu_0 = 0.4999$ in Section 5.2 and for its fixed-design reanalysis; in the compressible examples Poisson's ratio is fixed at ν_0 , and only Young's modulus varies with density according to (4.3).

The element compliance operator $\mathcal{A}(\bar{\rho}_e)$ is evaluated using the interpolated material parameters $E(\bar{\rho}_e)$ and $\nu(\bar{\rho}_e)$.

4.2 State Equations and Complementary Energy Compliance Formulation

We consider the minimum compliance problem under homogeneous displacement boundary conditions and design-independent external loads. Mixed finite elements take the stress and the displacement simultaneously as state variables and introduce the material distribution through the compliance block A of the saddle-point system (see [12]). In the Hu–Zhang discrete framework the total stress is obtained directly from the state equations, so we adopt a complementary-energy objective expressed in terms of the stress, which keeps the evaluation of the objective consistent with the primary variable of the mixed finite element method.

For the continuous linear elasticity problem, taking the test function equal to the solution in the weak form and using the constitutive relation shows that the external-load compliance equals twice the strain energy and also twice the complementary strain energy. Minimizing the compliance and minimizing the complementary strain energy therefore share the same minimizers. To keep the same scale as the external work used in displacement-based methods, we define the discrete objective without the factor $1/2$:

$$C(\rho, \sigma_h) = \int_{\Omega} \mathcal{A}(\bar{\rho}) \sigma_h : \sigma_h dx = \sum_{e=1}^{N_e} \mathbf{s}_e^T \mathbf{A}_e(\bar{\rho}_e) \mathbf{s}_e, \quad (4.5)$$

where $\bar{\rho} = \bar{\rho}(\rho)$ is the physical density obtained by the filtering and projection of Section 4.1, $\mathbf{s}_e$ is the coefficient vector of the total stress $\sigma_h|_{T_e}$ with respect to the local stress basis of the element, including the contribution of the traction lifting, and $\mathbf{A}_e$ is the corresponding local compliance matrix with (a, b) entry $\int_{T_e} \mathcal{A}(\bar{\rho}_e) \boldsymbol{\phi}_{e,b} : \boldsymbol{\phi}_{e,a} dx$, where $\{\boldsymbol{\phi}_{e,a}\}_{a=1}^{n_e}$ is a basis of $\mathbb{P}_k(T_e; \mathbb{S})$. Equation (4.5) is twice the discrete complementary strain energy and can be evaluated directly by element-wise integration and summation.

The state constraint is expressed in terms of the homogeneous stress unknowns of Section 3.5, and the total stress is reconstructed as $\sigma_h = \sigma_{0,h} + \sigma_{g,h}$. The objective of this section and the stress

constraints introduced later are both evaluated with this total stress σ_h . For a fixed mesh and the fixed stabilization scale of Section 3.3, $A(\bar{\rho})$ depends on the design variables through $E(\bar{\rho}_e)$ and $\nu(\bar{\rho}_e)$, whereas B and C remain unchanged. The complementary-energy-based minimum compliance model then reads

$$\begin{aligned}
\min_{\rho} \quad & C(\rho, \sigma_h) = \sum_{e=1}^{N_e} s_e^T A_e(\bar{\rho}_e) s_e \\
\text{s.t.} \quad & \begin{bmatrix} A(\bar{\rho}) & B \\ B^T & -C \end{bmatrix} \begin{bmatrix} s_0 \\ u \end{bmatrix} = \begin{bmatrix} f_\sigma(\bar{\rho}) \\ f_u \end{bmatrix}, \\
& \frac{\sum_{e=1}^{N_e} v_e \bar{\rho}_e}{\sum_{e=1}^{N_e} v_e} \leq \bar{V}, \\
& 0 \leq \rho_e \leq 1, \quad e = 1, \dots, N_e,
\end{aligned} \tag{4.6}$$

where $v_e = |T_e|$ is the element volume and $\bar{V} \in (0, 1)$ is the upper bound on the material volume fraction. The right-hand side is given by (3.14); under homogeneous displacement conditions its u_D term and ℓ_h^D vanish. Let s_g denote the coefficient vector of $\sigma_{g,h}$ with respect to the global basis of Σ_h^k ; extending s_0 by zero at the constrained degrees of freedom on Γ_N , the global coefficient vector of the total stress is $s = s_0 + s_g$, and s_e is its sub-vector associated with element T_e . The lifting $\sigma_{g,h}$ is design-independent, but the component $-\int_{\Omega} \mathcal{A}(\bar{\rho}) \sigma_{g,h} : \Phi_i dx$ of f_σ still depends on the material distribution, and the sensitivity analysis must account for this dependence.

This model evaluates the compliance directly from the total stress and the local compliance matrices, without constructing the displacement-form Schur complement $B^T A(\bar{\rho})^{-1} B + C$ and its design derivative. Under homogeneous displacement boundary conditions and design-independent loads, the mixed variational relations allow the implicit stress derivative to be eliminated from the objective sensitivity when no stabilization is present. The low-order stabilized formulation uses the same objective (4.5), but this simplification no longer applies. Its sensitivities are therefore evaluated using the general adjoint formula in Section 4.5.

4.3 Stress-Constrained Topology Optimization with a Relaxed Stress Threshold

We consider the minimization of the total structural volume subject to local stress constraints [13, 22], and impose a local von Mises stress constraint on every element of the domain:

$$g_e(\rho, \sigma_h) = \frac{\sigma_{\text{vm}}(\sigma_{h,e})}{\bar{\sigma}} - \eta(\bar{\rho}_e) \leq 0, \quad e = 1, \dots, N_e, \tag{4.7}$$

where $\bar{\sigma}$ is the allowable stress of the solid material and $\sigma_{h,e}$ is the apparent stress at the centroid of element T_e . To compare the two finite element discretizations, both use the apparent stress and the same ϵ -relaxed constraint: LFEM recovers the apparent stress $\mathcal{A}(\bar{\rho}_e)^{-1} \epsilon(u_h)$ from the displacement gradient and the interpolated constitutive law, whereas HZMFEM evaluates the reconstructed total stress $\sigma_h = \sigma_{0,h} + \sigma_{g,h}$. The two share the same volume objective, threshold

function, and evaluation points, and differ only in the source of the stress. Under plane stress conditions ($\sigma_{zz} = \sigma_{xz} = \sigma_{yz} = 0$), the von Mises stress is

$$\sigma_{\text{vm}}(\sigma_{h,e}) = \sqrt{\sigma_{xx}^2 - \sigma_{xx}\sigma_{yy} + \sigma_{yy}^2 + 3\sigma_{xy}^2},$$

where $\sigma_{xx}, \sigma_{yy}, \sigma_{xy}$ are the components of $\sigma_{h,e}$; for plane strain and three-dimensional problems the corresponding von Mises expression is used and the remaining derivation is unchanged.

If the allowable-stress condition is imposed directly on the solid-material stress, the constraint does not vanish as the element density tends to zero; the feasible set then contains degenerate lower-dimensional branches, giving rise to singular optima that gradient-based methods can hardly reach [20, 29, 35]. Following the ϵ -relaxation and the density-dependent stress limits used in mixed finite element formulations [13, 19, 22], we adopt the following threshold with a positive lower bound:

$$\eta(\bar{\rho}_e) = m_E(\bar{\rho}_e) + \epsilon(1 - m_E(\bar{\rho}_e)), \quad m_E(\bar{\rho}_e) = \frac{E(\bar{\rho}_e)}{E_0}, \quad (4.8)$$

where m_E is the relative stiffness, $E(\bar{\rho}_e)$ is given by (4.3), and $\epsilon \in (0, 1)$ is the relaxation parameter. Let σ_h^{app} denote the apparent stress field, with element centroid value $\sigma_e^{\text{app}} := \sigma_{h,e}$, and define the solid-material stress $\sigma_e^{\text{solid}} := \sigma_e^{\text{app}} / m_E(\bar{\rho}_e)$. When Poisson's ratio is held fixed, $\mathcal{A}(\bar{\rho}_e)^{-1} = m_E(\bar{\rho}_e) \mathcal{A}(1)^{-1}$, so σ_e^{solid} is the stress carried by the solid material under the same strain, which in the displacement-based method reads $\mathcal{A}(1)^{-1} \epsilon(u_h)$. By the first-order homogeneity of σ_{vm} , $\sigma_{\text{vm}}(\sigma_e^{\text{app}}) = m_E \sigma_{\text{vm}}(\sigma_e^{\text{solid}})$, so for $m_E > 0$ the constraint (4.7) is equivalent to $\sigma_{\text{vm}}(\sigma_e^{\text{solid}}) / \bar{\sigma} \leq 1 + \epsilon(1/m_E - 1)$, which relaxes the solid-material stress limit in regions of low stiffness. Hence (4.7) is an equivalent rewriting of the ϵ -relaxation: avoiding the explicit division by the density removes the numerical ill-conditioning as $m_E \rightarrow 0$, while the relaxation itself still comes from $\epsilon > 0$.

The threshold satisfies $\eta(1) = 1$, preserving the allowable-stress condition for the solid material. At $\bar{\rho}_e = 0$, it remains positive: $\eta(0) = \epsilon + (1 - \epsilon)E_{\min}/E_0$. The apparent stress is scaled by the small residual stiffness ratio, $\sigma_e^{\text{app}} = (E_{\min}/E_0)\sigma_e^{\text{solid}}$. Thus, the constraint becomes inactive when the corresponding apparent von Mises stress is below the relaxed limit $\eta(0)\bar{\sigma}$. We use a fixed relaxation parameter $\epsilon = 10^{-3}$ and do not apply ϵ -continuation.

Accordingly, the volume minimization problem with local stress constraints under the Hu-Zhang discretization reads

$$\begin{aligned} \min_{\rho} \quad & f_V(\rho) = \frac{\sum_{e=1}^{N_e} v_e \bar{\rho}_e}{\sum_{e=1}^{N_e} v_e} \\ \text{s.t.} \quad & \begin{bmatrix} A(\bar{\rho}) & B \\ B^T & -C \end{bmatrix} \begin{bmatrix} s_0 \\ u \end{bmatrix} = \begin{bmatrix} f_\sigma(\bar{\rho}) \\ f_u \end{bmatrix}, \\ & g_e(\rho, \sigma_h) = \frac{\sigma_{\text{vm}}(\sigma_{h,e})}{\bar{\sigma}} - \eta(\bar{\rho}_e) \leq 0, \quad e = 1, \dots, N_e, \\ & 0 \leq \rho_e \leq 1, \quad e = 1, \dots, N_e, \end{aligned} \quad (4.9)$$

where the physical density is determined as in Section 4.1, and the state equations and the reconstruction of the total stress follow Section 4.2. Since the volume of the design domain is fixed,

minimizing the volume fraction f_V is equivalent to minimizing the total structural volume. The LFEM counterpart replaces the state equations by $K(\bar{\rho})\mathbf{u} = \mathbf{f}$ and recovers the apparent stress through the interpolated constitutive law; the remaining components are identical to (4.9).

4.4 Augmented Lagrangian Formulation

Problem (4.9) imposes one local stress constraint on every element, so the number of constraints grows with the mesh size, and handling such a large number of constraints directly increases the cost of solving the optimization subproblems. We therefore follow the augmented Lagrangian method (ALM) of Senhora et al. [33] for local stress constraints: the local constraints are incorporated into the objective through augmentation terms and normalized by the number of constraints, giving the scalar augmented Lagrangian objective

$$\mathcal{L}(\boldsymbol{\rho}, \boldsymbol{\sigma}_h) = f_V(\boldsymbol{\rho}) + \frac{1}{N_e} \sum_{e=1}^{N_e} \mathcal{P}_e(g_e, \lambda_e, \mu_e), \quad (4.10)$$

where f_V is the volume fraction objective of (4.9). Since each element carries one local constraint, the number of constraints is N_e , and the factor $1/N_e$ mitigates the imbalance in magnitude between the augmentation terms and the objective as the number of constraints increases. For the local constraints defined in (4.7), we adopt the following form of the augmentation term [23]. Let $h_e = \max(g_e, -\lambda_e/\mu_e)$; the local augmentation term $\mathcal{P}_e = \lambda_e h_e + \frac{\mu_e}{2} h_e^2$ can then be written equivalently as

$$\mathcal{P}_e = \begin{cases} \lambda_e g_e + \frac{\mu_e}{2} g_e^2, & \text{if } g_e > -\frac{\lambda_e}{\mu_e}, \\ -\frac{\lambda_e^2}{2\mu_e}, & \text{if } g_e \leq -\frac{\lambda_e}{\mu_e}, \end{cases} \quad (4.11)$$

where $\lambda_e \geq 0$ is the Lagrange multiplier and $\mu_e > 0$ is the penalty parameter. In the k -th outer iteration, the multipliers $\lambda^{(k)}$ and penalty parameters $\mu^{(k)}$ are held fixed, and the augmented subproblem subject only to the bounds on the design variables is solved:

$$\begin{aligned} \min_{\boldsymbol{\rho}} \quad & \mathcal{L}^{(k)}(\boldsymbol{\rho}) = f_V(\boldsymbol{\rho}) + \frac{1}{N_e} \sum_{e=1}^{N_e} \mathcal{P}_e(g_e(\boldsymbol{\rho}, \boldsymbol{\sigma}_h(\boldsymbol{\rho})), \lambda_e^{(k)}, \mu_e^{(k)}), \\ \text{s.t.} \quad & 0 \leq \rho_e \leq 1, \quad e = 1, \dots, N_e, \end{aligned} \quad (4.12)$$

where $\boldsymbol{\sigma}_h(\boldsymbol{\rho})$ is obtained by solving the state equations in (4.9) and reconstructing the total stress. The state variables have been eliminated from this subproblem, so only the bounds on the design variables remain explicit. After the subproblem is solved approximately, the outer multipliers are updated with the constraint values g_e at the approximate solution and projected onto the interval $[0, \lambda_{\max}]$:

$$\lambda_e^{(k+1)} = \min\left(\lambda_{\max}, \max\left(0, \lambda_e^{(k)} + \mu_e^{(k)} g_e\right)\right), \quad (4.13)$$

where $\lambda_{\max}$ is the upper bound on the multipliers, following the safeguarded augmented Lagrangian scheme of Andreani et al. [2]. If the constraint of an element cannot be satisfied throughout the iterations, its multiplier stops growing at $\lambda_{\max}$ and the final design may be infeasible in

that element, so constraint satisfaction of the final design must be re-checked element by element. The subproblem (4.12) is solved approximately in the inner loop by the method of moving asymptotes (MMA) [34]; the update rule for the penalty parameters, the final constraint check, and the complete solution procedure are given in Section 4.6.

4.5 Consistent Adjoint Sensitivity Analysis

This section derives the sensitivities of the objective functions with respect to the physical density, accounting for the design dependence of the right-hand side $f_\sigma(\bar{\rho})$. The derivation is carried out with the mesh, the multipliers and penalty parameters, the projection parameters, and the stabilization scale held fixed, and with a design-independent traction lifting; the state matrix is assumed nonsingular and the objective differentiable. Throughout, $\partial/\partial\bar{\rho}_e$ denotes the partial derivative with the state variables held fixed, and $d/d\bar{\rho}_e$ the total derivative that includes the state response.

For the augmented Lagrangian objective $J = \mathcal{L}$ or the complementary-energy objective $J = C$, we introduce the adjoint variables (z_σ, z_u) and solve the adjoint system

$$[A(\bar{\rho}) \quad B \quad B^\top \quad -C] [z_\sigma \quad z_u] = [\nabla_{s_0} J \quad \mathbf{0}]. \quad (4.14)$$

Since the state matrix is symmetric, the adjoint system shares its coefficient matrix with the state equations. Let $\zeta_h = \sum_i (z_\sigma)_i \Phi_i$ denote the adjoint stress. By the chain rule applied to the material interpolation (4.3)–(4.4),

$$\mathcal{A}'_e := \frac{\partial \mathcal{A}}{\partial \bar{\rho}_e} = \frac{\partial \mathcal{A}}{\partial E} \frac{dE}{d\bar{\rho}_e} + \frac{\partial \mathcal{A}}{\partial \nu} \frac{d\nu}{d\bar{\rho}_e},$$

where the second term vanishes when Poisson's ratio is fixed.

For the augmented Lagrangian objective $\mathcal{L}$, let $\omega_e = \partial \mathcal{P}_e / \partial g_e = \max(0, \lambda_e + \mu_e g_e)$. The adjoint load and the total derivative with respect to the physical density are, respectively,

$$(\nabla_{s_0} \mathcal{L})_i = \frac{1}{N_e \bar{\sigma}} \sum_{m=1}^{N_e} \omega_m \frac{\partial \sigma_{vm}(\sigma_{h,m})}{\partial s_{0,i}},$$

$$\frac{d\mathcal{L}}{d\bar{\rho}_e} = \frac{v_e}{\sum_m v_m} - \frac{\omega_e}{N_e} \eta'(\bar{\rho}_e) - \int_{T_e} \mathcal{A}'_e \sigma_h : \zeta_h dx, \quad \eta'(\bar{\rho}_e) = (1 - \epsilon) m'_E(\bar{\rho}_e), \quad (4.15)$$

where the integral term combines the design derivatives of the state matrix and of the right-hand side $f_\sigma(\bar{\rho})$ and is therefore expressed in terms of the reconstructed total stress σ_h . The von Mises stress and its derivative are evaluated at the element centroids as defined in (4.7), and the gradient formula holds wherever the objective is differentiable.

For the complementary-energy objective C , differentiating (4.5) gives the adjoint load $(\nabla_{s_0} C)_i = 2 \int_\Omega \mathcal{A}(\bar{\rho}) \sigma_h : \Phi_i dx$. With the corresponding ζ_h obtained from (4.14),

$$\frac{dC}{d\bar{\rho}_e} = \int_{T_e} \mathcal{A}'_e \sigma_h : (\sigma_h - \zeta_h) dx. \quad (4.16)$$

In the absence of stabilization, under homogeneous displacement conditions and design-independent loads, the adjoint system has the explicit solution $\mathbf{z}_\sigma = \mathbf{0}$, $\mathbf{z}_u = -2\mathbf{u}$, i.e., $\boldsymbol{\zeta}_h = \mathbf{0}$, so no separate adjoint solve is required and (4.16) reduces to

$$\frac{dC}{d\bar{\rho}_e} = \mathbf{s}_e^\top \frac{\partial A_e}{\partial \bar{\rho}_e} \mathbf{s}_e, \quad c_h = 0, \quad (4.17)$$

where $\mathbf{s}_e$ and A_e are defined in (4.5); the low-order stabilized scheme ($1 \leq k \leq d$, $c_h \neq 0$) uses (4.16).

The LFEM counterpart uses the displacement state equation $\mathbf{K}(\bar{\boldsymbol{\rho}})\mathbf{u} = \mathbf{f}$. For the augmented objective $\mathcal{L}(\boldsymbol{\rho}, \mathbf{u})$ under design-independent loads, solving the adjoint equation $\mathbf{K}(\bar{\boldsymbol{\rho}})\mathbf{z} = \nabla_{\mathbf{u}}\mathcal{L}$, which shares the symmetric stiffness matrix with the state equation, gives

$$\frac{d\mathcal{L}}{d\bar{\rho}_e} = \left. \frac{\partial \mathcal{L}}{\partial \bar{\rho}_e} \right|_{\mathbf{u}} - \mathbf{z}^\top \frac{\partial \mathbf{K}}{\partial \bar{\rho}_e} \mathbf{u}, \quad (4.18)$$

where the partial derivative at fixed state consists of three parts: $v_e / \sum_m v_m$ from the volume objective, $-\omega_e \eta'(\bar{\rho}_e) / N_e$ from the relaxed threshold, and the explicit density dependence of the apparent stress $\mathcal{A}(\bar{\rho}_e)^{-1} \boldsymbol{\varepsilon}(\mathbf{u}_h)$ through the interpolated constitutive law; the last term is absent from (4.15) because the stress is a state variable in HZMFEM. For the compliance objective, LFEM takes the external work $C = \mathbf{f}^\top \mathbf{u}$, whose adjoint solution is $\mathbf{z} = \mathbf{u}$ [8], and the gradient is

$$\frac{dC}{d\bar{\rho}_e} = -\mathbf{u}^\top \frac{\partial \mathbf{K}}{\partial \bar{\rho}_e} \mathbf{u}. \quad (4.19)$$

Finally, for $J \in \{\mathcal{L}, C\}$, the gradient is transferred back to the design variables through the projection (4.2) and the filter (4.1):

$$\frac{dJ}{d\rho_j} = \sum_{e: j \in \mathcal{N}_e} \frac{dJ}{d\bar{\rho}_e} P'_{\beta, \eta_p}(\tilde{\rho}_e) \frac{w_{ej} v_j}{\sum_{m \in \mathcal{N}_e} w_{em} v_m}, \quad (4.20)$$

where the derivative of the projection is

$$P'_{\beta, \eta_p}(\tilde{\rho}_e) = \frac{\beta \left\{ 1 - \tanh^2[\beta(\tilde{\rho}_e - \eta_p)] \right\}}{\tanh(\beta \eta_p) + \tanh[\beta(1 - \eta_p)]}.$$

The volume fraction f_V follows the same rule with $df_V / d\bar{\rho}_e = v_e / \sum_m v_m$; when the projection is disabled (identity map), $P'_{\beta, \eta_p} = 1$.

4.6 Optimization Algorithms and Numerical Workflows

Problem (4.6) is solved by MMA or by the optimality criteria (OC) method [8]; problem (4.9) is solved by the combination of an outer ALM loop and an inner MMA loop described in Section 4.4, denoted ALM–MMA.

4.6.1 Single-Loop Algorithm for Compliance Minimization

Algorithm 4.1 summarizes the single-loop procedure for problem (4.6), in which only the density filter is applied and the state equations are solved by either HZMFEM or LFEM.

Algorithm 4.1 Single-loop optimization procedure based on MMA or OC

Input: Initial design $\rho^{(0)}$; filter parameters; optimizer type (MMA or OC) and its parameters; volume fraction bound $\bar{V}$; move limit m ; stopping parameters ε_ρ , $N_{\max}$.

Output: Final design variables ρ^* with the corresponding physical density $\bar{\rho}$, compliance C , and volume fraction f_V .

- 1: Set $n=0$, $\Delta_\rho^{(0)} = \infty$; initialize the MMA asymptotes and design history if MMA is used.
 - 2: Filter $\rho^{(0)}$; solve the state equations; compute $C^{(0)}$ and $f_V^{(0)}$.
 - 3: **while** the stopping criterion is not met and $n < N_{\max}$ **do**
 - 4: Compute $dC/d\rho_e$ and $df_V/d\rho_e$.
 - 5: Update $\rho^{(n+1)}$ by MMA or OC under the volume constraint, the design bounds, and the move limit.
 - 6: Filter; solve the state equations; compute $C^{(n+1)}$ and $f_V^{(n+1)}$.
 - 7: Compute $\Delta_\rho^{(n+1)} = \|\rho^{(n+1)} - \rho^{(n)}\|_\infty$.
 - 8: Set $n \leftarrow n+1$; the stopping criterion is met when $\Delta_\rho^{(n)} \leq \varepsilon_\rho$ and $f_V^{(n)} \leq \bar{V}$.
 - 9: **end while**
 - 10: **return** $\rho^* = \rho^{(n)}$ with the corresponding results.
-

In step 5 of Algorithm 4.1, MMA constructs and solves an approximate subproblem for the objective and volume constraint using moving asymptotes. Alternatively, OC uses the damped multiplicative update of Andreassen et al. [3], with the multiplier determined by bisection to satisfy the volume constraint. Both updates respect the design-variable bounds and the move limit m . The volume is evaluated using the filtered physical density. The move limit m and the OC damping exponent ξ are specified for each example.

4.6.2 Nested Two-Loop Algorithm for Stress-Constrained Volume Minimization

Algorithm 4.2 summarizes the nested two-loop procedure for problem (4.9), in which the density filter and the projection are applied and the state equations are solved by either HZMFEM or LFEM; $\rho^{k,j}$ denotes the design after j inner updates within outer step k .

The outer ALM loop updates the multipliers by (4.13), and the penalty factors grow by the factor γ_μ and are capped at $\mu_{\max}$ as in step 12 of Algorithm 4.2; since all elements start from the same penalty factor and are updated simultaneously, $\mu_e^{(k)}$ takes the same value for all e , and neither the multipliers nor the penalty factors are reset when the projection parameters change. The inner loop approximately solves the augmented subproblem by MMA with a fixed number of $N_{\text{in}}^{\max}$ steps. In step 10, the maximum local constraint value on the criterion set and the mean

Algorithm 4.2 Nested two-loop optimization procedure based on ALM–MMA

Input: Initial design $\rho^{0,0}$; filter, projection (with continuation schedule), and MMA parameters; ALM parameters $\mu_0, \gamma_\mu, \mu_{\max}, \lambda_{\max}$; criterion set $\mathcal{E}_{\text{acc}}$; stopping parameters $\delta_\rho, \delta_g, N_{\text{hold}}, N_{\text{out}}^{\max}, N_{\text{in}}^{\max}$.

Output: Final design ρ^* with its physical density $\bar{\rho}$, stress field σ_h , volume fraction f_V , and maximum local constraint values $g_{\max}$ and $g_{\max,\text{all}}$.

- 1: Set $k = 0, q = 0, \lambda_e^{(0)} = 0, \mu_e^{(0)} = \mu_0$; initialize the projection parameter β and the MMA asymptotes and design history.
- 2: Filter and project $\rho^{0,0}$; solve the state equations; compute f_V and g_e .
- 3: **while** $q < N_{\text{hold}}$ and $k < N_{\text{out}}^{\max}$ **do**
- 4: Store $\rho^{k,0}$; fix $\lambda^{(k)}, \mu^{(k)}, \beta$; compute $\mathcal{L}^{(k)}$; set $j = 0$.
- 5: **while** $j < N_{\text{in}}^{\max}$ **do**
- 6: Solve the adjoint equations; compute $d\mathcal{L}^{(k)}/d\rho_e$ through the filter and projection chain rule.
- 7: Update $\rho^{k,j+1}$ by MMA within the design bounds and the move limit.
- 8: Filter and project; solve the state equations; compute $f_V, g_e, \mathcal{L}^{(k)}$; set $j \leftarrow j + 1$.
- 9: **end while**
- 10: Set $\rho^{k+1,0} = \rho^{k,j}$; compute $g_{\max}^{k+1}$ and $\Delta_{\rho,\text{out}}^{k+1}$.
- 11: If the continuation is complete, the projection was unchanged in this outer step, $g_{\max}^{k+1} \leq \delta_g$, and $\Delta_{\rho,\text{out}}^{k+1} < \delta_\rho$, set $q \leftarrow q + 1$; otherwise set $q = 0$.
- 12: If $q = N_{\text{hold}}$ or $k + 1 = N_{\text{out}}^{\max}$, set $k \leftarrow k + 1$ and exit; otherwise update and truncate $\lambda_e^{(k+1)}$ by (4.13), set $\mu_e^{(k+1)} = \min(\gamma_\mu \mu_e^{(k)}, \mu_{\max})$, $k \leftarrow k + 1$.
- 13: Update the projection parameters by the schedule; if changed, recompute the physical density, state, and constraints, and reset the MMA history and q .
- 14: **end while**
- 15: **return** $\rho^* = \rho^{k,0}$; re-solve the state equations; report $f_V, g_{\max}$, and $g_{\max,\text{all}}$.

design change between the beginning and the end of the outer step are defined as

$$g_{\max}^{k+1} = \max_{e \in \mathcal{E}_{\text{acc}}} g_e, \quad \Delta_{\rho,\text{out}}^{k+1} = \frac{1}{N_e} \sum_{e=1}^{N_e} |\rho_e^{k+1,0} - \rho_e^{k,0}|,$$

where the latter follows the form of Giraldo-Londoño and Paulino [23]; $g_{\max}$ is the signed maximum local constraint value, $g_{\max,\text{all}} = \max_{1 \leq e \leq N_e} g_e$ in step 15 is taken over all elements, and all local constraints are strictly satisfied when $g_{\max,\text{all}} < 0$.

5 Numerical Results

In this section, we first verify the convergence of the discretizations of different orders through a manufactured solution, and then examine the performance of the proposed method in compli-

ance minimization, nearly incompressible material design, and local stress-constrained problems through three classes of TO examples. All examples are implemented within the open-source finite element library FEALPy and the TO platform SOPTX.

5.1 Convergence Rate Verification via Manufactured Solutions

An isotropic plane-strain manufactured solution problem is constructed on the unit square $\Omega = (0,1)^2$ with Lamé parameters $\lambda = 1$ and $\mu = 0.5$. The exact displacement field is taken as

$$\mathbf{u}(x,y) = \begin{bmatrix} \sin(\pi x)\sin(\pi y) \\ \sin(\pi x)\sin(\pi y) \end{bmatrix},$$

the exact stress is given by $\boldsymbol{\sigma} = \mathcal{A}^{-1}\boldsymbol{\varepsilon}(\mathbf{u})$, and the body force is taken as $\mathbf{b} = -\operatorname{div}\boldsymbol{\sigma}$. Mixed boundary conditions are adopted: the exact displacement vanishes on $\Gamma_D = \{x=0\} \cup \{y=0\}$, where a homogeneous displacement condition is imposed, and the exact traction $\mathbf{g} = \boldsymbol{\sigma}\mathbf{n}$ is prescribed on $\Gamma_N = \{x=1\} \cup \{y=1\}$. The mesh is a uniform right-triangle partition obtained by dividing each coordinate direction into n_x equal segments with diagonals alternating in a checker-board pattern, so that each corner of the domain belongs to exactly two triangles; the two-element corner relaxation of Section 3.4 is activated at all four corners, keeping the same discrete setting as in the subsequent TO examples. Five refinement levels $n_x = 4, 8, 16, 32, 64$ are considered, with maximum element diameter $h = \sqrt{2}/n_x$. In the tables, the total number of degrees of freedom is the sum of the stress and displacement degrees of freedom, and the observed convergence order is computed from the error ratio of two consecutive levels as $\log_2(e_h/e_{h/2})$.

5.1.1 Higher-Order Convergence

The high-order schemes ($k=3,4$) are solved without stabilization; the errors and observed convergence orders are listed in Table 1.

Table 1 shows that, for $k=3$ and $k=4$, the observed convergence orders of the L^2 displacement error and the $H(\operatorname{div})$ stress error both approach k , while that of the L^2 stress error approaches $k+1$, in agreement with the optimal-order estimates (3.3)–(3.4). These estimates are established for the standard problem with pure displacement boundary conditions; the present example indicates that the optimal orders are preserved under mixed boundary conditions and corner relaxation.

5.1.2 Lower-Order Stabilized Convergence

The low-order schemes ($k=1,2$) employ the matrix jump stabilization (3.7) with coefficient $\alpha = \mu/L_0^2 = 0.5$ ($L_0 = 1$); the errors and observed convergence orders are listed in Table 2.

It can be observed from Table 2 that, for $k=1$ and $k=2$, the observed convergence orders of the L^2 displacement error and the $H(\operatorname{div})$ stress error both approach k , numerically attaining the optimal order $\mathcal{O}(h^k)$ guaranteed by (3.10). Standard norms are used in the table: for fixed λ the weighted stress norm is equivalent to the standard $H(\operatorname{div})$ norm, and on the displacement side $\|\mathbf{u} - \mathbf{u}_h\|_0 \leq \|\mathbf{u} - \mathbf{u}_h\|_{0,c}$, so the orders in (3.10) apply directly. The observed order of the L^2 stress error is about 1.5 for $k=1$ and about 3 for $k=2$, both satisfying the $\mathcal{O}(h^k)$ estimate

Table 1: Convergence errors and observed orders of the high-order Hu–Zhang mixed finite elements ($k=3,4$) for the manufactured solution.

k	n_x	Total DOFs	$\ u - u_h\ _0$	Order	$\ \sigma - \sigma_h\ _0$	Order	$\ \sigma - \sigma_h\ _{H(\text{div})}$	Order
3	4	975	3.0644×10^{-3}	–	4.0519×10^{-3}	–	8.8146×10^{-2}	–
	8	3767	3.8850×10^{-4}	2.98	2.3422×10^{-4}	4.11	1.1180×10^{-2}	2.98
	16	14823	4.8746×10^{-5}	2.99	1.4118×10^{-5}	4.05	1.4027×10^{-3}	2.99
	32	58823	6.0991×10^{-6}	3.00	8.6802×10^{-7}	4.02	1.7550×10^{-4}	3.00
	64	234375	7.6256×10^{-7}	3.00	5.3837×10^{-8}	4.01	2.1943×10^{-5}	3.00
4	4	1631	2.6778×10^{-4}	–	2.7107×10^{-4}	–	7.7075×10^{-3}	–
	8	6359	1.6969×10^{-5}	3.98	8.9171×10^{-6}	4.93	4.8836×10^{-4}	3.98
	16	25127	1.0643×10^{-6}	3.99	2.8610×10^{-7}	4.96	3.0627×10^{-5}	4.00
	32	99911	6.6579×10^{-8}	4.00	9.0347×10^{-9}	4.98	1.9158×10^{-6}	4.00
	64	398471	4.1621×10^{-9}	4.00	2.8345×10^{-10}	4.99	1.1976×10^{-7}	4.00

Table 2: Convergence errors and observed orders of the low-order jump-stabilized Hu–Zhang mixed finite elements ($k=1,2$) for the manufactured solution.

k	n_x	Total DOFs	$\ u - u_h\ _0$	Order	$\ \sigma - \sigma_h\ _0$	Order	$\ \sigma - \sigma_h\ _{H(\text{div})}$	Order
1	4	143	4.3496×10^{-1}	–	7.9802×10^{-1}	–	5.4342×10^0	–
	8	503	2.5631×10^{-1}	0.76	2.5985×10^{-1}	1.62	2.7370×10^0	0.99
	16	1895	1.2408×10^{-1}	1.05	8.8099×10^{-2}	1.56	1.3696×10^0	1.00
	32	7367	6.0154×10^{-2}	1.04	3.0532×10^{-2}	1.53	6.8511×10^{-1}	1.00
	64	29063	2.9637×10^{-2}	1.02	1.0721×10^{-2}	1.51	3.4265×10^{-1}	1.00
2	4	479	2.7735×10^{-2}	–	5.4679×10^{-2}	–	7.9819×10^{-1}	–
	8	1815	7.0044×10^{-3}	1.99	7.4165×10^{-3}	2.88	2.0267×10^{-1}	1.98
	16	7079	1.7574×10^{-3}	1.99	9.6884×10^{-4}	2.94	5.0875×10^{-2}	1.99
	32	27975	4.3978×10^{-4}	2.00	1.2343×10^{-4}	2.97	1.2733×10^{-2}	2.00
	64	111239	1.0997×10^{-4}	2.00	1.5545×10^{-5}	2.99	3.1843×10^{-3}	2.00

contained in (3.10); for $k=2$ the optimal interpolation order $k+1$ is attained, whereas for $k=1$ the order is higher than the theoretical one but remains suboptimal, consistent with the fact that the L^2 stress estimate in (3.10) is itself suboptimal [16]. That analysis addresses the pure displacement boundary problem; the present example involves a traction boundary, the jump penalty is imposed only on interior faces and the displacement boundary, and the non-homogeneous traction data are enforced through the discrete lifting. The observed results indicate that the order conclusions remain valid in this setting. These results provide the numerical basis for the response analysis and the evaluation of stress-based objectives and constraints in the subsequent TO.

5.2 Benchmark Topology Optimization Cases

In all examples, LFEM and HZMFEM share the same density mapping and material interpolation; the cases in which the Poisson ratio interpolation is activated are stated in Section 4.1. All HZMFEM runs activate the two-element corner relaxation of Section 3.4 at the four corners of the rectangular design domain. HZMFEM with $k=2$ uniformly uses the jump stabilization of Section 3.3 with a fixed coefficient, which is not updated with the density in either the optimization or the reanalysis. LFEM uses continuous Lagrange displacement approximations of degree p , solves $K(\bar{\rho})\mathbf{u} = \mathbf{f}$, and evaluates the compliance as the external work $\mathbf{f}^\top \mathbf{u}$; the local stress constraints are computed from the apparent stress defined in Section 4.3. The compliance of HZMFEM is evaluated as the complementary energy (4.5); it and the external work are two discrete approximations of the same continuous quantity, and their discrete values for the same design are in general not equal (Section 4.2). Sections 5.2.1 and 5.2.2 use the single-loop procedure of Algorithm 4.1 with MMA and OC updates, respectively, and Section 5.2.3 uses the ALM–MMA procedure of Algorithm 4.2.

5.2.1 Compliance Minimization of Clamped Beam

We consider the compliance minimization benchmark of a clamped beam (Figure 3). The design domain measures $160\text{mm} \times 20\text{mm}$ under plane stress, with material parameters $E_0 = 30\text{MPa}$ and $\nu_0 = 0.4$ and volume fraction bound $\bar{V} = 0.4$. Since HZMFEM imposes the traction condition as an essential condition on the normal trace of the stress (Section 2.3), a concentrated force cannot be applied directly. The point load $P = 3\text{N}$ at the midpoint of the bottom edge is therefore converted into an equivalent uniform traction of intensity $\bar{t} = P/l = 3\text{N/mm}$ over a contact zone of characteristic width $l = 1\text{mm}$, and is applied through a projection onto the continuous $\mathbb{P}_1$ boundary trace space. Both discretizations thus use the same load distribution and the same discrete traction data.

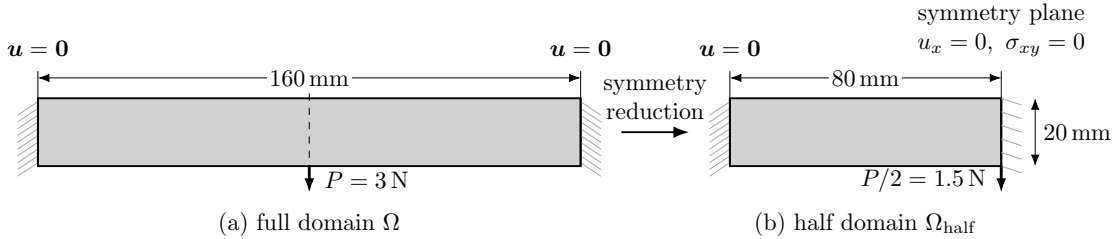


Figure 3: Geometry, loading and symmetry boundary conditions of the clamped beam: (a) full design domain; (b) left-half computational domain.

Exploiting the symmetry of the geometry and the loading, the computations are carried out on the left half of the domain ($80\text{mm} \times 20\text{mm}$), with the conditions $u_x = 0$ and $\sigma_{xy} = 0$ on the symmetry plane $x = 80\text{mm}$. LFEM imposes $u_x = 0$ strongly on the horizontal displacement DOFs. In HZMFEM, $u_x = 0$ is a natural condition, and $\sigma_{xy} = 0$ is imposed strongly by setting the tangential component of the stress normal trace on this edge to zero while leaving the normal component free.

The resultant load carried by the half domain is $P/2 = 1.5\text{N}$; in what follows the compliance is reported for the full structure, i.e., twice the value computed on the half domain.

The same uniform right-triangle partition with checkerboard-alternating diagonals as in Section 5.1 is used with 80×20 cells ($N_e = 3200$ elements on the half domain, equivalent to 6400 elements on the full domain), and the density filter radius is $r_{\min} = 2.4\text{mm}$. Both discretizations use the MMA update of Algorithm 4.1, starting from the uniform initial density $\rho_e^{(0)} = \bar{V}$, with move limit $m = 0.2$, design-change tolerance $\varepsilon_\rho = 10^{-2}$, bounds $0 \leq \rho_e \leq 1$, void modulus $E_{\min} = 10^{-9}\text{MPa}$ and penalization exponent $p_E = 3$. LFEM is run with $p = 2, 3, 4$ and HZMFEM with $k = 2, 3, 4$, where $k = 2$ uses the jump stabilization and $k = 3, 4$ the native high-order scheme. The final topologies are compared in Figure 4 (mirrored to the full domain), and the convergence histories are shown in Figure 5.

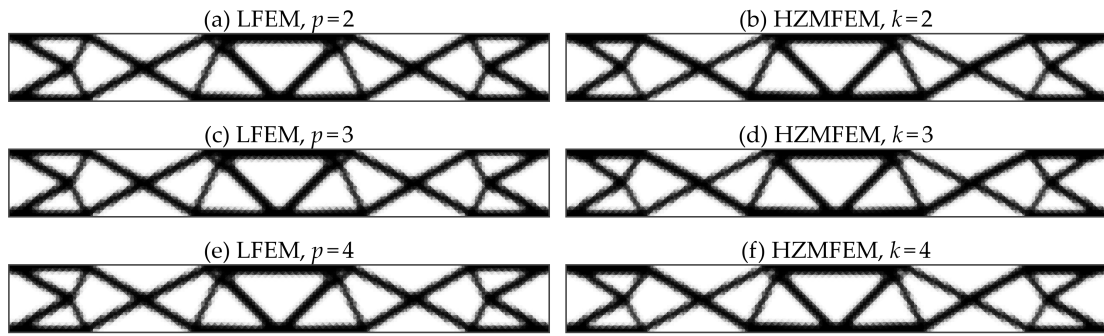


Figure 4: Final topologies of the clamped beam obtained with MMA (left column: LFEM, p is the displacement degree; right column: HZMFEM, k is the stress degree).

Figure 4 shows that both methods produce layouts whose main load path consists of crossing diagonal bars, with differences in the connections and thicknesses of individual bars. The three LFEM runs with $p = 2, 3, 4$ reach the design-change tolerance after 230, 230 and 254 iterations, and the three HZMFEM runs with $k = 2, 3, 4$ after 287, 301 and 342 iterations; the final volume fractions are all close to 0.4 and satisfy the volume constraint (Figure 5). The final full-beam compliance of each run evaluated with its own discretization (the optimized compliance) is listed in the third column of Table 3 (in $\text{N} \cdot \text{mm}$, likewise below): 31.73, 31.83 and 31.85 for LFEM with $p = 2, 3, 4$, and 32.76, 32.11 and 31.91 for HZMFEM with $k = 2, 3, 4$.

The differences among these optimized compliances cannot be read directly as differences among the final designs. The displacement method evaluates the compliance as the external work and the mixed method as the complementary energy (4.5), so the two give different discrete values even for the same design; moreover, each run solves its own optimization problem independently, and the final designs are not identical. To separate the two factors, the final physical densities of the six runs are frozen, the state equations are solved once with each of LFEM $p = 3, 4$ and HZMFEM $k = 3, 4$, and the compliance is evaluated with the corresponding functional; the results are listed in the last four columns of Table 3.

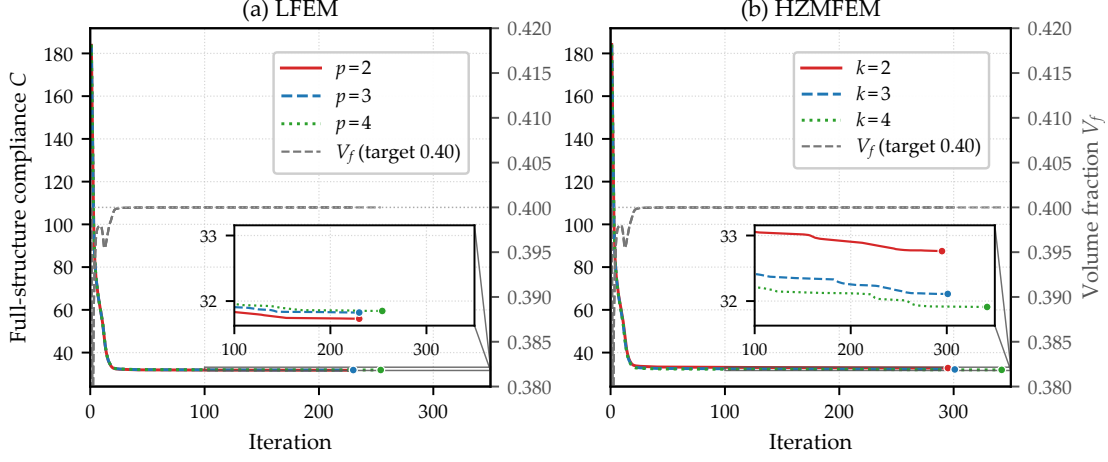


Figure 5: Optimization histories of the clamped beam with MMA (colored curves: full-structure compliance; gray curves: volume fraction; end markers: final compliance of each run, i.e., the optimized compliance in Table 3; inset: zoom of the late iterations).

Table 3: Optimized compliance and uniform reanalysis compliance of the six final designs of the clamped beam (full structure, in $\text{N}\cdot\text{mm}$; p : displacement degree of LFEM, k : stress degree of HZMFEM).

Optimized design	Iterations	Optimized compliance	Reanalysis compliance			
			$p=3$	$p=4$	$k=3$	$k=4$
LFEM $p=2$	230	31.73	31.83	31.85	32.38	32.13
LFEM $p=3$	230	31.83	31.83	31.85	32.38	32.13
LFEM $p=4$	254	31.85	31.83	31.85	32.38	32.13
HZMFEM $k=2$	287	32.76	31.60	31.63	32.13	31.88
HZMFEM $k=3$	301	32.11	31.57	31.59	32.11	31.86
HZMFEM $k=4$	342	31.91	31.61	31.64	32.17	31.91

Within each reanalysis column, the compliances of the three LFEM designs differ by at most 0.01% and those of the three HZMFEM designs by at most 0.20%, and the maximum relative difference among the six designs is about 0.82% (LFEM columns) and 0.86% (HZMFEM columns), far below the 3.26% spread among the six optimized compliances. On the other hand, for each frozen design the LFEM compliance increases with p while the HZMFEM compliance decreases with k , and the relative difference between the two functionals at the same order shrinks from 1.66%–1.78% at order 3 to 0.82%–0.89% at order 4, so the two discrete approximations approach each other from opposite sides. The differences among the optimized compliances therefore stem mainly from the objective functionals and the discretizations rather than from the final designs. This agrees with the observation of Bruggi [11] that truly mixed and displacement-based elements

on regular grids yield almost identical optimal layouts for energy-type TO problems. This example shows that HZMFEM attains load-carrying layouts and final compliances close to those of LFEM in the classical compliance minimization problem, providing numerical support for the correctness of the proposed TO pipeline.

5.2.2 Nearly Incompressible Topology Optimization (2D Bearing)

The 2D bearing benchmark ($120\text{mm} \times 40\text{mm}$, plane strain, Figure 6) is used to examine the resistance to volumetric locking in the nearly incompressible limit. The bottom edge is fully clamped ($u_x = u_y = 0$), a uniform downward traction of intensity $t_0 = 8 \times 10^{-2} \text{N/mm}$ is applied on the top edge, and the left and right edges are traction-free. The material parameters are $E_0 = 1 \text{MPa}$, with $\nu_0 = 0.3$ for the compressible reference group and $\nu_0 = 0.4999$ for the nearly incompressible group; the void modulus is $E_{\min} = 10^{-9} \text{MPa}$ and $p_E = 3$. The Poisson's ratio is kept fixed in the compressible group, whereas in the nearly incompressible group it is interpolated by (4.4) with $\nu_{\text{void}} = 0.3$ and $p_v = 1$.

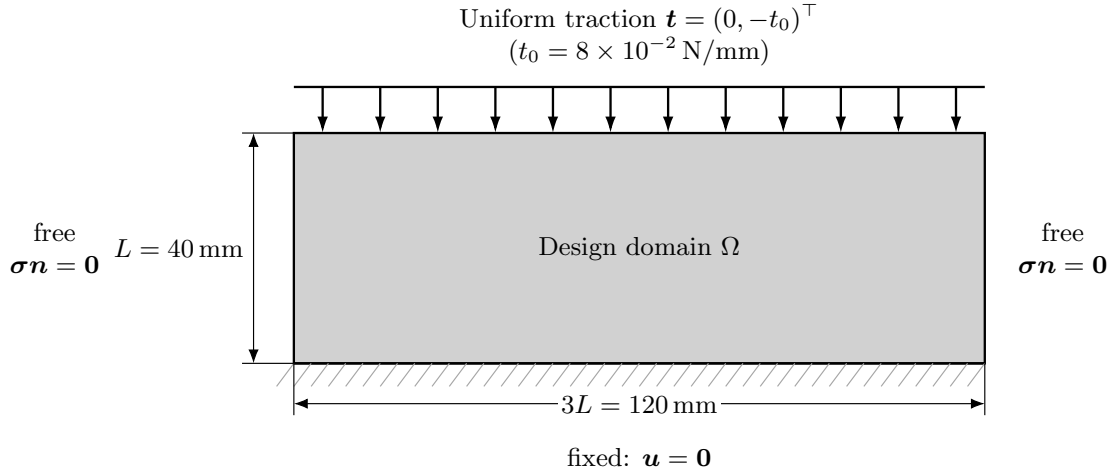


Figure 6: Geometry and boundary conditions of the 2D bearing.

The design domain is discretized by a 120×40 structured triangular mesh (9600 elements) whose diagonals are mirrored between the left and right halves, consistent with the left-right symmetry of the problem. The volume fraction bound is $\bar{V} = 0.35$, the density filter radius is $r_{\min} = 2.0 \text{mm}$, and the initial density is uniform, $\rho_e^{(0)} = \bar{V}$. For each material group the problem is solved with LFEM $p = 1, 2$ and HZMFEM $k = 2$ (the latter with the jump stabilization of Section 3.3), giving six optimization runs in total. All runs use the OC update of Algorithm 4.1 with move limit $m = 0.2$, damping exponent $\xi = 0.5$ and design-change tolerance $\varepsilon_\rho = 10^{-2}$, and all reach the stopping criterion.

Before the optimization, volumetric locking is examined on the fully solid domain ($\rho_e \equiv 1$). For $\nu_0 = 0.3$ and 0.4999 the mesh is refined from 30×10 to 240×80 ; LFEM $p = 1, 2$ and HZMFEM $k = 2$ are computed on four levels, and HZMFEM $k = 4$, which needs no stabilization, on

the first three. For each Poisson's ratio the reference value is the Richardson-extrapolated limit of the first three HZMFEM $k=4$ results [32] (Figure 7). The limits extrapolated from the other sequences differ from this reference by no more than 0.06%, so deviations below this level (the two finest levels of LFEM $p=2$ in the compressible group) are not interpreted quantitatively. On the 120×40 mesh, the absolute compliance deviation of LFEM $p=1$ grows from about 0.04% to 16.82% as the Poisson's ratio is increased, a clear sign of locking; those of HZMFEM $k=2$ and $k=4$ grow only from 0.66% and 0.30% to 0.96% and 0.42%, and for both Poisson's ratios they decrease monotonically under refinement with similar magnitudes, showing none of the accuracy degradation of the linear displacement element. The deviation of LFEM $p=2$ in the nearly incompressible group is about 0.13%, still below that of both mixed discretizations.

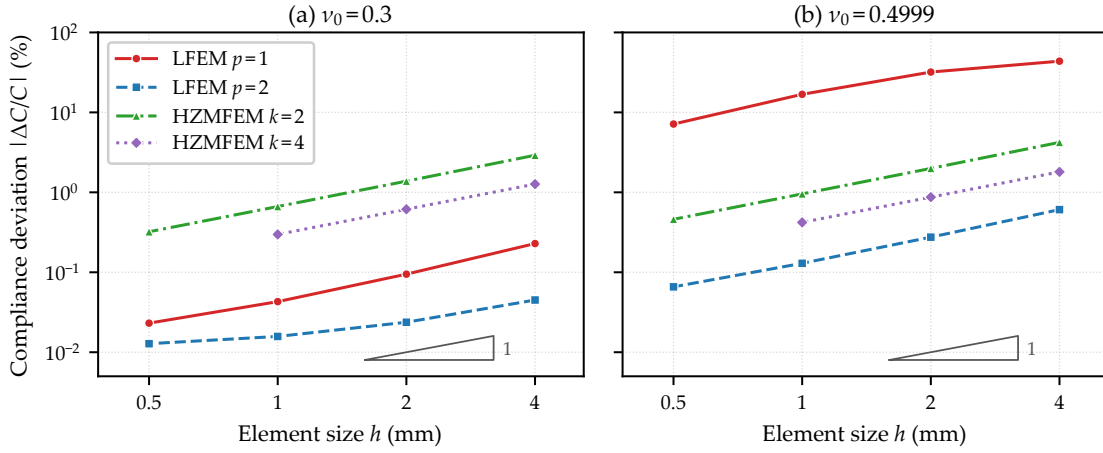


Figure 7: Absolute compliance deviation from the extrapolated reference on the fully solid domain ($\rho_e \equiv 1$): (a) $\nu_0 = 0.3$; (b) $\nu_0 = 0.4999$.

Figure 8 shows the final topologies of the six runs, all of which form a load path dominated by three arches. In the compressible group the three discretizations give similar layouts. In the nearly incompressible group, the arches of LFEM $p=1$ have more polygonal contours with sharper crowns and locally thicker members, whereas LFEM $p=2$ and HZMFEM $k=2$ retain smooth arch contours and give close layouts. Together with the marked accuracy degradation of the linear displacement element in Figure 7, these differences indicate that volumetric locking has affected its optimization result. In the nearly incompressible group LFEM $p=1$ exhibits a spurious over-stiffness due to locking, so its computed compliance is too low (optimized-compliance column of Table 4) and cannot be taken as evidence of a better design.

To compare the six final designs, the physical densities of each run are frozen and reanalyzed on the 120×40 mesh with LFEM $p=1,2$ and HZMFEM $k=2,4$; Table 4 lists the HZMFEM $k=4$ compliance together with the relative deviations of the other discretizations from it. Under the common HZMFEM $k=4$ reanalysis, the compliances of the three compressible designs differ by at most about 1.2%; in the nearly incompressible group, the design obtained with LFEM $p=1$

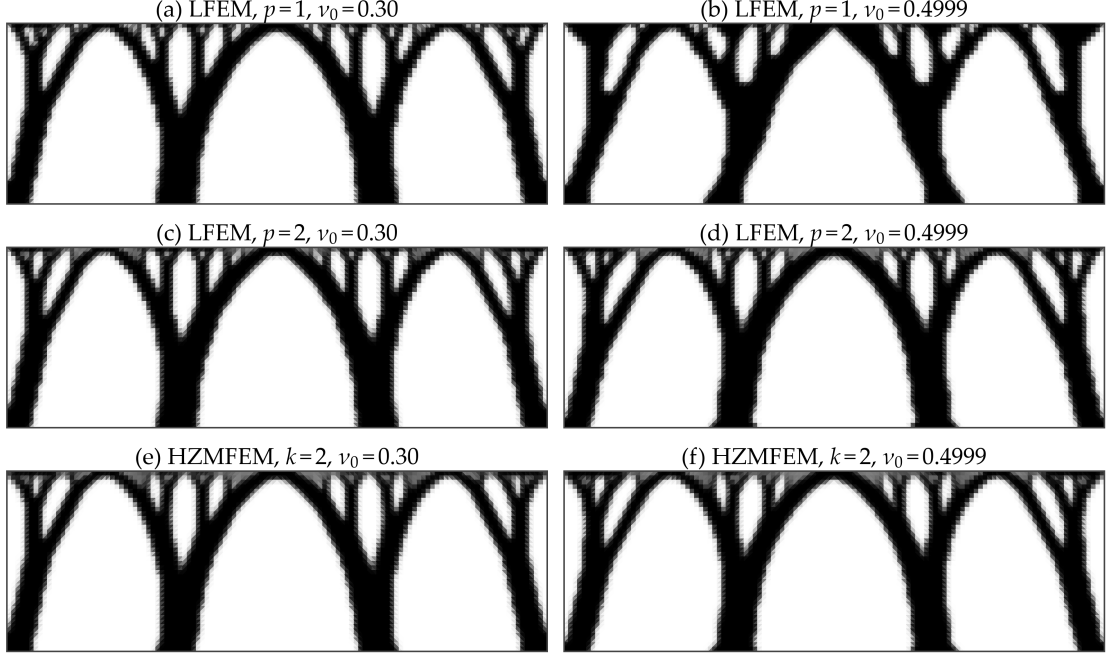


Figure 8: Final topologies of the 2D bearing (left column: $\nu_0 = 0.30$; right column: $\nu_0 = 0.4999$; from top to bottom: LFEM $p = 1$, LFEM $p = 2$, HZMFEM $k = 2$).

has a compliance about 24.0% higher than that obtained with HZMFEM $k = 2$, whereas LFEM $p = 2$ and HZMFEM $k = 2$ differ by only about 0.5%. For the same frozen design, the LFEM $p = 1$ compliance in the nearly incompressible group is about 17%–38% below the reference value, which further shows that locking overestimates its stiffness. Combining Figures 7 and 8, the Hu–Zhang element effectively mitigates the influence of the locking of the linear displacement element on the optimization result, and its designs perform close to those of the quadratic displacement element.

5.2.3 Local Stress-Constrained Cantilever Beam

The local stress-constrained cantilever beam (Figure 9) has a design domain of $80\text{mm} \times 40\text{mm}$ in plane stress, clamped along the left edge. A vertical resultant $P = 400\text{N}$ at the midpoint of the right edge is applied as a uniform traction $\bar{t} = P/l$ over a contact width $l = 6\text{mm}$ whose end points coincide with mesh nodes. The displacement method imposes this load through the boundary integral and the mixed method through the normal trace of the stress; the neighborhoods of the contact end points are treated as passive solid regions as described below. The solid material is an aluminum alloy with $E_0 = 70000\text{MPa}$, $\nu_0 = 0.25$ and allowable stress $\bar{\sigma} = 180\text{MPa}$, and the void modulus is $E_{\min} = 10^{-9}E_0$. Since the load is design-independent and the displacement boundary conditions are homogeneous, the stresses do not depend on the scale of the elastic moduli; the computations therefore use $E_0^{\text{calc}} = 1\text{MPa}$, with $E_{\min}$ and the stabilization coefficient of the stabi-

Table 4: Iterations, optimized compliance and HZMFEM $k=4$ reanalysis compliance of the six final designs of the 2D bearing (in N·mm), and relative deviations of the other reanalysis discretizations from the HZMFEM $k=4$ value.

ν_0	Optimized design	Iterations	Optimized compliance	Reanalysis $k=4$	Deviation from $k=4$		
					$p=1$	$p=2$	$k=2$
0.30	LFEM $p=1$	281	119.33	125.83	-5.2%	-1.9%	+4.9%
0.30	LFEM $p=2$	351	122.73	124.64	-4.0%	-1.5%	+4.0%
0.30	HZMFEM $k=2$	464	127.68	124.30	-3.5%	-1.2%	+2.7%
0.4999	LFEM $p=1$	372	76.43	123.49	-38.1%	-2.6%	+4.9%
0.4999	LFEM $p=2$	245	98.06	100.01	-17.8%	-1.9%	+4.0%
0.4999	HZMFEM $k=2$	343	102.37	99.56	-16.8%	-1.6%	+2.8%

lized $k=2$ scheme used later in this section scaled proportionally, while the load and $\bar{\sigma}$ keep their physical values. The penalization exponent is $p_E=3.5$, the domain is discretized by an 80×40 checkerboard triangular mesh with alternating diagonals (6400 elements), the density filter radius is $r_{\min}=6.0\text{mm}$, and the initial density is $\rho_e^{(0)}=0.5$.

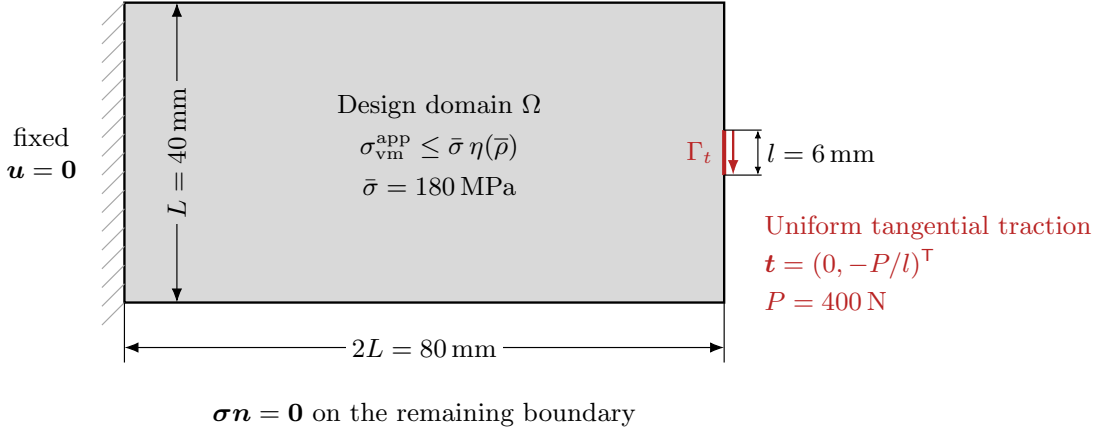


Figure 9: Design domain, local load and boundary conditions of the 2D cantilever beam.

Both methods use the same mesh, load and optimization parameters. LFEM uses $p=3$ and HZMFEM uses $k=3$, the latter being a native high-order scheme that needs no jump stabilization. The filtered density is passed through the tanh projection (4.2) with $\eta_p=0.5$, and β is increased from 1 by 1 every 5 outer iterations up to 10. The local stress constraints follow the model of Section 4.3 with $\epsilon=10^{-3}$ held fixed throughout. A passive solid region Ω_{pad} is defined as the neighborhoods of radius $r_{\text{pad}}=1.5\text{mm}$ around the two end points of the contact width (16 elements for both discretizations, 0.25% of the design domain area). No local stress constraint is imposed there, i.e. the region is left out of the stress-constraint term of the augmented Lagrangian, as for the

elements around concentrated loads in Bruggi and Venini [13]. Its physical density is set to 1 after filtering and projection and the corresponding derivative of the density mapping is set to zero, so that the region is excluded from the optimization as in Holmberg et al. [25]. Its volume is counted in the total volume fraction. The criterion set is $\mathcal{E}_{\text{acc}} = \{e \notin \Omega_{\text{pad}} : \bar{\rho}_e \geq 0.5\}$; the elements with $\bar{\rho}_e < 0.5$ still carry the relaxed stress constraint but do not enter the stopping test. The ALM–MMA procedure of Algorithm 4.2 is used with $\mu_0 = 50$, $\gamma_\mu = 1.1$, $\mu_{\text{max}} = 10^4$, $\lambda_{\text{max}} = 3000$, move limit $m = 0.15$, a minimum asymptote gap of 10^{-4} , $N_{\text{out}}^{\text{max}} = 200$ and $N_{\text{in}}^{\text{max}} = 5$. The stopping parameters are $\delta_\rho = 0.002$, $\delta_g = 0.005$ and $N_{\text{hold}} = 3$.

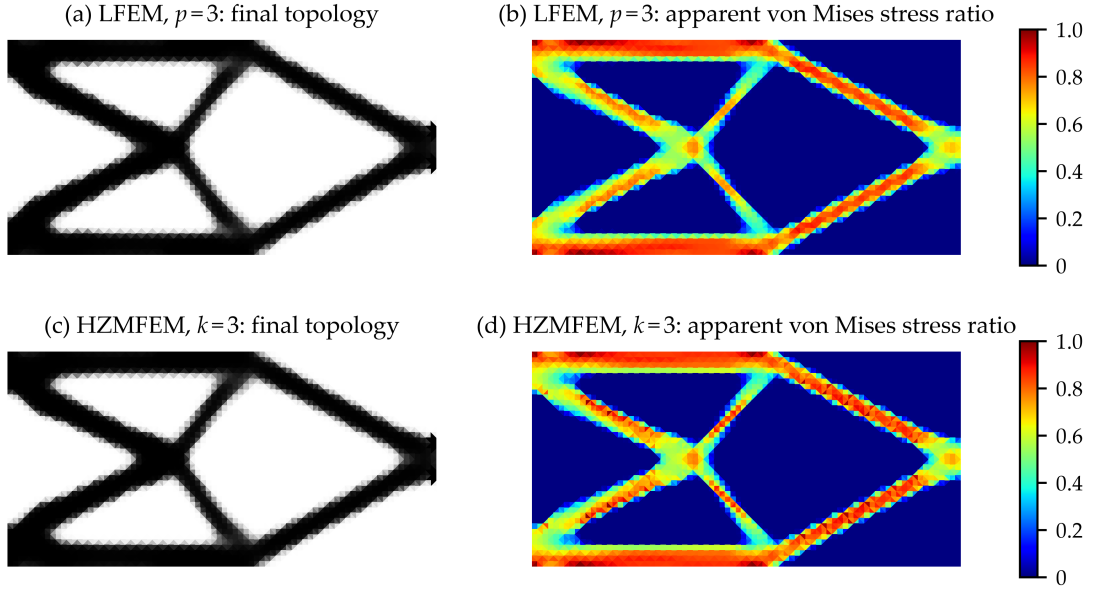


Figure 10: Final topologies and apparent von Mises stress ratio of the cantilever beam with cubic discretizations (LFEM: $p = 3$; HZMFEM: $k = 3$).

Figure 10 shows that the two discretizations give similar truss-like topologies: the main load path consists of the upper and lower chords and the internal crossing diagonals, the high-stress regions are distributed along this path, and both the layout and the stress distribution are close between the two discretizations.

Figure 11(a,c) shows the histories of the volume fraction and of the maximum local constraint value on the criterion set $\mathcal{E}_{\text{acc}}$, each constraint being evaluated with its own discretization. LFEM and HZMFEM reach the stopping criterion of Algorithm 4.2 after 248 and 256 MMA updates, respectively; the final f_V are 34.39% and 35.01%, and the final g_{max} are 3.92×10^{-3} and 2.82×10^{-3} , both below $\delta_g = 0.005$. Figure 11(b,d) shows the ratios $\sigma_i / \bar{\sigma}$ ($\sigma_1 \geq \sigma_2$) of the apparent principal stresses at the element centroids to the allowable stress for the elements in $\mathcal{E}_{\text{acc}}$, colored by the apparent von Mises stress ratio. The stress points lie almost entirely in the quadrant $\sigma_1 \geq 0 \geq \sigma_2$: part of them cluster along the two axes, i.e. in uniaxial tension or compression, in line with

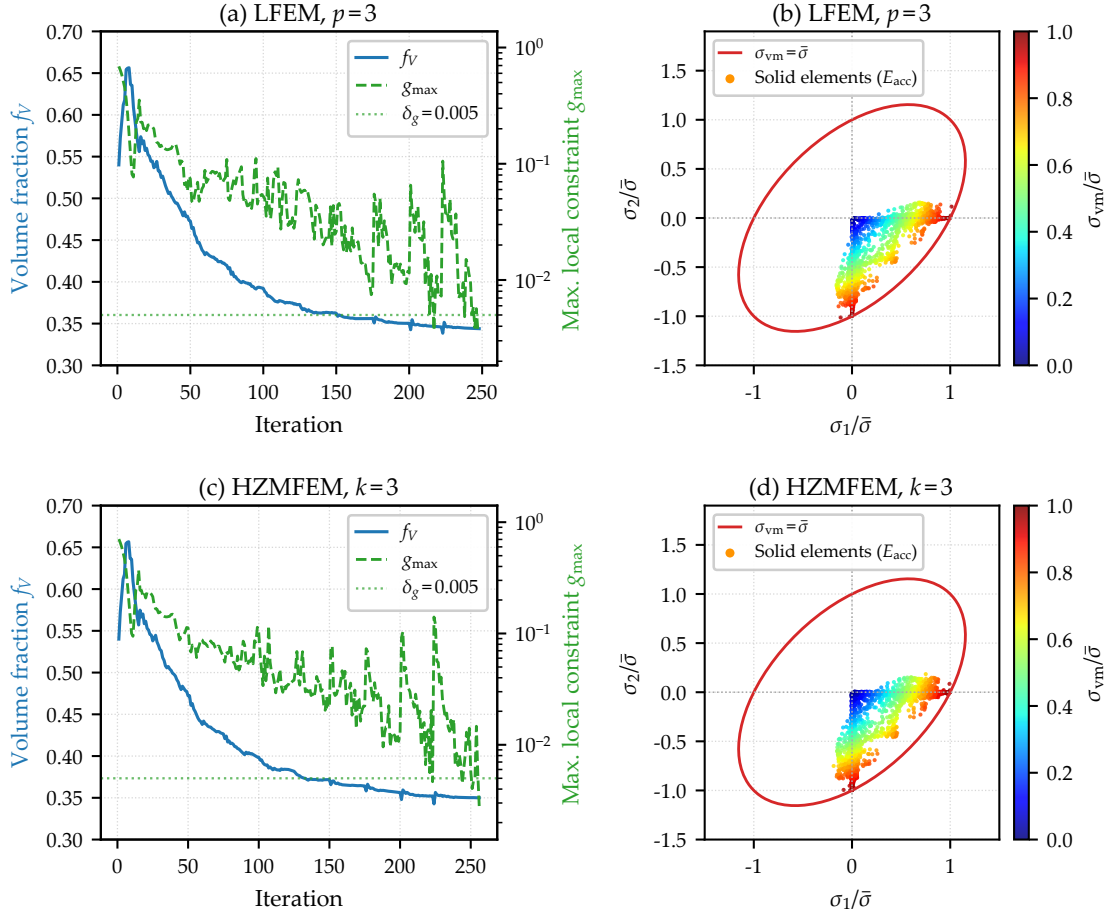


Figure 11: Convergence histories and element stress distributions in the principal stress plane for the cantilever beam with cubic discretizations ((a,b) LFEM $p = 3$; (c,d) HZMFEM $k = 3$); $g_{\max}$ and the stress points are evaluated on the criterion set $\mathcal{E}_{acc}$.

the axial load-carrying of the members of the truss-like layout, while the rest spread between the axes up to the yield ellipse, i.e. in biaxial tension–compression states dominated by shear.

With the same settings as in Figure 10, the problem is also solved with the stabilized scheme $k=2$ and the native scheme $k=4$; the results are shown in Figure 12. Together with $k=3$, the three orders take 250, 256 and 258 MMA updates, the final f_v are 34.70%, 35.01% and 35.30%, and the final $g_{\max}$ on the criterion set $\mathcal{E}_{acc}$ are 1.96×10^{-3} , 2.82×10^{-3} and 4.92×10^{-3} , all below δ_g and all satisfying the same design-change criterion. The three orders give similar main load-carrying layouts, showing that the optimization procedure applies to both the low-order stabilized and the high-order native discretizations considered.

To examine the stress fields of the two discretizations further, we compare their inter-element

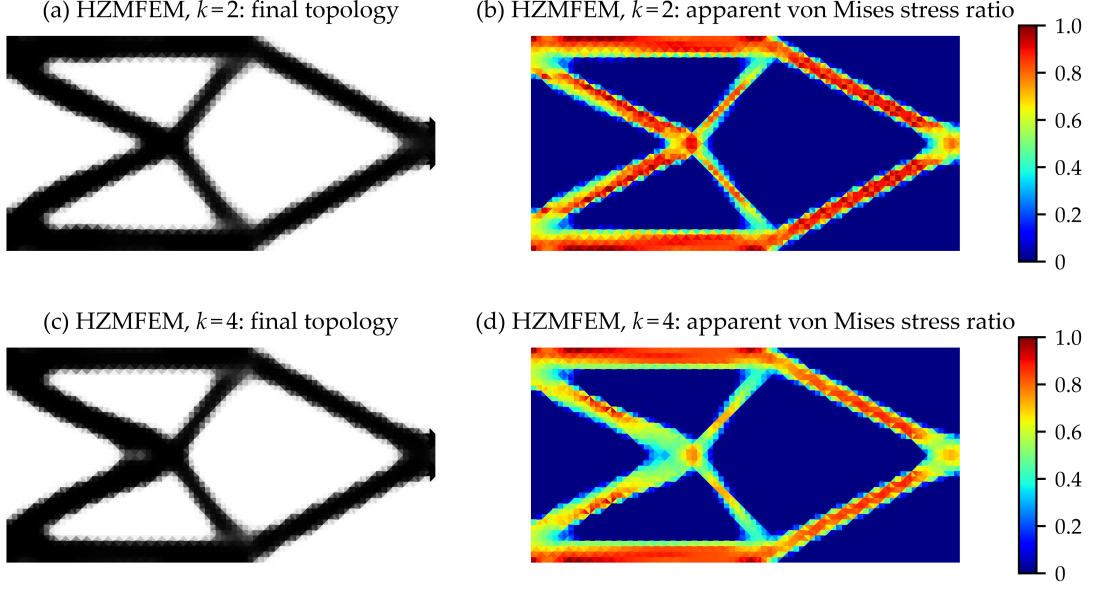


Figure 12: Final topologies and apparent von Mises stress ratio of the cantilever beam with Hu-Zhang mixed discretizations ((a,b) $k=2$; (c,d) $k=4$).

normal traction continuity. For HZMFEM, g_e is taken directly from the primary unknown $\sigma_h \in H(\text{div}, \Omega; \mathbf{S})$, whose normal traction is single-valued across elements; for LFEM, the stress is recovered element by element from the displacement gradient, and the traction is discontinuous on interior faces. To quantify this difference, the six final designs of LFEM $p=2,3,4$ and HZMFEM $k=2,3,4$ (those for $p=2,4$ are obtained separately with the settings and stopping criterion of Figure 10) are re-solved for the state equations with the discretization used in their optimization. On an interior face F we define the normalized root-mean-square traction jump A_F , and take the maximum over the interior faces of an element as the element indicator A_e :

$$A_F = \frac{1}{\bar{\sigma}} \left(\frac{1}{|F|} \int_F \| [\![\sigma_h^{\text{app}} \mathbf{n}]\!] \|^2 ds \right)^{1/2}, \quad A_e = \max_{F \subset \partial T_e \cap \Omega} A_F. \quad (5.1)$$

The statistics are restricted to the solid band ($\bar{\rho}_e > 0.9$ and $e \notin \Omega_{\text{pad}}$). In Figure 13, A_e of the three HZMFEM orders is at round-off level (at most 2.2×10^{-16}); for LFEM $p=2,3,4$ the medians of A_e are 2.4×10^{-2} , 1.6×10^{-2} and 1.1×10^{-2} and the 95th percentiles are 0.13, 0.086 and 0.072, decreasing with the order but remaining clearly non-zero, with the larger jumps concentrated at the member edges. This reflects the difference between the two discretizations in inter-element equilibrium: the Hu-Zhang stress field preserves normal traction continuity, whereas a stress recovered element by element from the displacement gradient does not satisfy this continuity in general. The lines δ_g and $4\delta_g$ in the figure serve only as numerical scale references and are not acceptance thresholds for the traction jump.

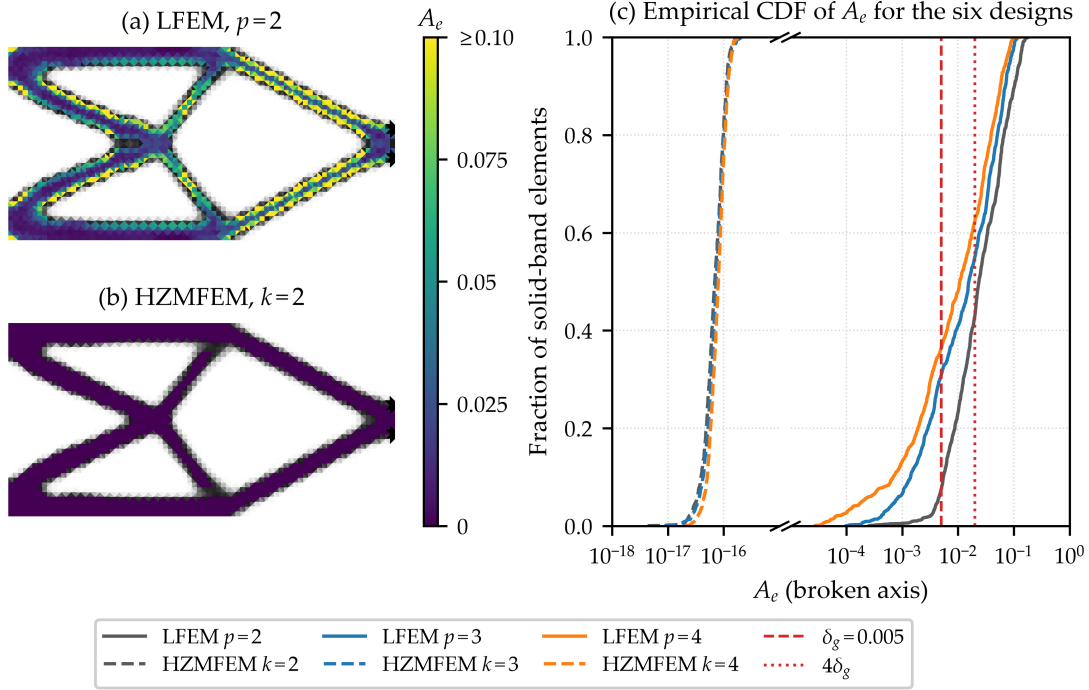


Figure 13: Element-wise traction jump A_e on the solid band of the final designs: spatial distribution for (a) LFEM $p=2$ and (b) HZMFEM $k=2$ (shared color scale, values above the upper bound are clipped); (c) empirical cumulative distributions of the six designs on a broken horizontal axis, the left part for HZMFEM $k=2-4$ (dashed) and the right part for LFEM $p=2-4$ (solid), with vertical lines at δ_g and $4\delta_g$.

6 Concluding Remarks

6.1 Key Conclusions

This paper has established a high-order Hu–Zhang mixed finite element framework for density-based topology optimization. The main conclusions are as follows:

1. **High-order Hu–Zhang TO framework:** Within the Hellinger–Reissner framework with non-homogeneous traction boundaries, the essential traction condition is imposed through a design-independent lifting field; the lifting term makes the discrete right-hand side depend on the material distribution, and the consistent adjoint sensitivities of the complementary-energy objective and of the local stress constraints account for this dependence. Through the management of normal-trace degrees of freedom, local corner relaxation and jump stabilization for lower orders, the stabilized formulations with $k \leq d$ and the native formulations with $k \geq d+1$ run within the same optimization workflow. The observed convergence orders in the manufactured-solution tests agree with the theoretical estimates; in the clamped

beam benchmark, the two discretizations yield similar load-carrying layouts and close compliances under a common reanalysis.

2. **Topology optimization under nearly incompressible conditions:** The compliance bilinear form of the Hu–Zhang elements remains bounded as $\lambda \rightarrow \infty$; no λ -uniform discrete stability estimate is established in this paper, and the locking resistance is examined numerically. In the bearing benchmark, the Hu–Zhang elements exhibit similar fully solid compliance convergence trends for the two Poisson’s ratios, without the pronounced loss of accuracy observed for linear displacement elements; the common reanalysis shows that, in the nearly incompressible group, the designs obtained with Hu–Zhang elements have lower compliance than those obtained with linear displacement elements and are close to those obtained with quadratic displacement elements.
3. **Local stress-constrained TO driven by independent stresses:** The Hu–Zhang stress field belongs to $H(\text{div}, \Omega; \mathbb{S})$, so its normal traction is single-valued across elements in the trace sense. The local von Mises stress constraints are constructed directly from this primary stress variable and combined with a density-dependent relaxed threshold and the ALM–MMA procedure to solve the volume minimization problem, without recovering stresses from displacement gradients. In the cantilever benchmark, HZMFEM $k = 3$ and LFEM $p = 3$ yield similar layouts and stress distributions, HZMFEM $k = 2, 3, 4$ all reach the stopping criterion, with the constraints evaluated on the criterion set by each discretization; the element-wise traction jumps of the Hu–Zhang stress fields of all three orders are at round-off level, whereas those of the stresses recovered from displacement gradients are clearly non-zero.

6.2 Future Work and Outlook

The numerical study of this paper is restricted to two-dimensional small-strain linear elasticity. Future work will proceed along the following three directions:

1. **Three-dimensional extension and parallel solvers.** Extend the local corner relaxation and the treatment of boundary conditions to three-dimensional Hu–Zhang discretizations, and study the organization of stress degrees of freedom on edges and vertices where boundaries meet, so that prescribed traction conditions remain compatible with inter-element normal-trace continuity. To address the growth of unknowns for higher-order discretizations and the solution difficulties caused by material stiffness contrast, study block preconditioners and parallel solvers for the mixed saddle-point systems, and reuse solver information between successive optimization steps to reduce the computational cost.
2. **Error-driven mesh and order adaptivity.** Develop a posteriori error estimates for compliance and local stress constraints, and apply local mesh refinement and order adjustment according to the corresponding accuracy requirements, concentrating computational resources on stress concentration regions and material transition regions. Meanwhile, study the trans-

fer of design variables across mesh changes and the coordinated treatment of filtering and projection to keep the design length scale consistent.

3. **Finite deformation and nonlinear materials.** For nearly incompressible hyperelastic structures, develop stress–displacement mixed discretizations compatible with finite-deformation kinematics, together with the corresponding material interpolation and consistent adjoint sensitivities. To address the large deformation and mesh distortion in low-density regions, study material regularization and nonlinear solution strategies to improve the robustness of the optimization process, and assess stress accuracy, constraint satisfaction and computational cost through benchmarks at different load levels and material parameters.

Acknowledgments

References

- [1] S. Adams and B. Cockburn. A mixed finite element method for elasticity in three dimensions. *Journal of Scientific Computing*, 25:515–521, 2005.
- [2] R. Andreani, E. G. Birgin, J. M. Martínez, and M. L. Schuverdt. On augmented Lagrangian methods with general lower-level constraints. *SIAM Journal on Optimization*, 18(4):1286–1309, 2008.
- [3] E. Andreassen, A. Clausen, M. Schevenels, B. S. Lazarov, and O. Sigmund. Efficient topology optimization in MATLAB using 88 lines of code. *Structural and Multidisciplinary Optimization*, 43(1):1–16, 2011.
- [4] D. N. Arnold, R. S. Falk, and R. Winther. Mixed finite element methods for linear elasticity with weakly imposed symmetry. *Mathematics of Computation*, 76(260):1699–1723, 2007.
- [5] D. N. Arnold and R. Winther. Mixed finite elements for elasticity. *Numerische Mathematik*, 92(3):401–419, 2002.
- [6] I. Babuška and M. Suri. Locking effects in the finite element approximation of elasticity problems. *Numerische Mathematik*, 62(1):439–463, 1992.
- [7] M. P. Bendsøe. Optimal shape design as a material distribution problem. *Structural Optimization*, 1(4):193–202, 1989.
- [8] M. P. Bendsøe and O. Sigmund. *Topology Optimization: Theory, Methods, and Applications*. Springer, 2004.
- [9] D. Boffi, F. Brezzi, and M. Fortin. *Mixed Finite Element Methods and Applications*, volume 44 of *Springer Series in Computational Mathematics*. Springer, Berlin, Heidelberg, 2013.
- [10] B. Bourdin. Filters in topology optimization. *International Journal for Numerical Methods in Engineering*, 50(9):2143–2158, 2001.
- [11] M. Bruggi. Topology optimization with mixed finite elements on regular grids. *Computer Methods in Applied Mechanics and Engineering*, 305:133–153, 2016.
- [12] M. Bruggi and P. Venini. Topology optimization of incompressible media using mixed finite elements. *Computer Methods in Applied Mechanics and Engineering*, 196(33–34):3151–3164, 2007.
- [13] M. Bruggi and P. Venini. A mixed FEM approach to stress-constrained topology optimization. *International Journal for Numerical Methods in Engineering*, 73(12):1693–1714, 2008.
- [14] T. E. Bruns and D. A. Tortorelli. Topology optimization of non-linear elastic structures and compliant mechanisms. *Computer Methods in Applied Mechanics and Engineering*, 190(26–27):3443–3459, 2001.

- [15] C. Chen, L. Chen, X. Huang, and H. Wei. Geometric decomposition and efficient implementation of high order face and edge elements. *Communications in Computational Physics*, 35(4):1045–1072, 2024.
- [16] L. Chen, J. Hu, and X. Huang. Stabilized mixed finite element methods for linear elasticity on simplicial grids in $\mathbb{R}^n$. *Computational Methods in Applied Mathematics*, 17(1):17–31, 2017.
- [17] L. Chen, J. Hu, and X. Huang. Fast auxiliary space preconditioners for linear elasticity in mixed form. *Mathematics of Computation*, 87(312):1601–1633, 2018.
- [18] L. Chen, J. Hu, X. Huang, and H. Man. Residual-based a posteriori error estimates for symmetric conforming mixed finite elements for linear elasticity problems. *Science China Mathematics*, 61(6):973–992, 2018.
- [19] G. Cheng and X. Guo. ϵ -relaxed approach in structural topology optimization. *Structural Optimization*, 13(4):258–266, 1997.
- [20] G. Cheng and Z. Jiang. Study on topology optimization with stress constraints. *Engineering Optimization*, 20(2):129–148, 1992.
- [21] R. Codina, I. Castañar, and J. Baiges. Finite element approximation of stabilized mixed models in finite strain hyperelasticity involving displacements and stresses and/or pressure—an overview of alternatives. *International Journal for Numerical Methods in Engineering*, 125(18):e7540, 2024.
- [22] P. Duysinx and M. P. Bendsøe. Topology optimization of continuum structures with local stress constraints. *International Journal for Numerical Methods in Engineering*, 43(8):1453–1478, 1998.
- [23] O. Giraldo-Londoño and G. H. Paulino. PolyStress: a Matlab implementation for local stress-constrained topology optimization using the augmented Lagrangian method. *Structural and Multidisciplinary Optimization*, 63(4):2065–2097, 2021.
- [24] L. He, H. Wei, and T. Tian. SOPTX: A modular and extensible framework for topology optimization with multi-backend support. *Communications in Computational Physics*, 40(2):525–572, 2026.
- [25] E. Holmberg, B. Torstenfelt, and A. Klarbring. Stress constrained topology optimization. *Structural and Multidisciplinary Optimization*, 48(1):33–47, 2013.
- [26] J. Hu. Finite element approximations of symmetric tensors on simplicial grids in $\mathbb{R}^n$: the higher order case. *Journal of Computational Mathematics*, 33(3):283–296, 2015.
- [27] J. Hu and R. Ma. Partial relaxation of C^0 vertex continuity of stresses of conforming mixed finite elements for the elasticity problem. *Computational Methods in Applied Mathematics*, 21(1):89–108, 2021.
- [28] J. Hu and S. Zhang. A family of conforming mixed finite elements for linear elasticity on triangular grids. *arXiv preprint arXiv:1406.7457*, 2014.
- [29] U. Kirsch. On singular topologies in optimum structural design. *Structural Optimization*, 2(3):133–142, 1990.
- [30] C. Le, J. Norato, T. Bruns, C. Ha, and D. Tortorelli. Stress-based topology optimization for continua. *Structural and Multidisciplinary Optimization*, 41(4):605–620, 2010.
- [31] J. París, F. Navarrina, I. Colominas, and M. Casteleiro. Topology optimization of continuum structures with local and global stress constraints. *Structural and Multidisciplinary Optimization*, 39(4):419–437, 2009.
- [32] P. J. Roache. Perspective: a method for uniform reporting of grid refinement studies. *Journal of Fluids Engineering*, 116(3):405–413, 1994.
- [33] F. V. Senhora, O. Giraldo-Londoño, I. F. M. Menezes, and G. H. Paulino. Topology optimization with local stress constraints: a stress aggregation-free approach. *Structural and Multidisciplinary Optimization*, 62(4):1639–1668, 2020.
- [34] K. Svanberg. The method of moving asymptotes—a new method for structural optimization. *International Journal for Numerical Methods in Engineering*, 24(2):359–373, 1987.

- [35] G. Sved and Z. Ginos. Structural optimization under multiple loading. *International Journal of Mechanical Sciences*, 10(10):803–805, 1968.
- [36] F. Wang, B. S. Lazarov, and O. Sigmund. On projection methods, convergence and robust formulations in topology optimization. *Structural and Multidisciplinary Optimization*, 43(6):767–784, 2011.
- [37] Y. Zheng, H. Wei, Y. Huang, C. Chen, T. Tian, H. Liu, W. Wang, and L. He. FEALPy: A cross-platform intelligent numerical simulation engine. *Communications in Computational Physics*, 40(5):1676–1704, 2026.